ФЕДЕРАЛЬНОЕ ГОСУДАРСТВЕННОЕ АВТОНОМНОЕ
ОБРАЗОВАТЕЛЬНОЕ УЧРЕЖДЕНИЕ
ВЫСШЕГО ПРОФЕССИОНАЛЬНОГО ОБРАЗОВАНИЯ
«НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ УНИВЕРСИТЕТ
“ВЫСШАЯ ШКОЛА ЭКОНОМИКИ”»
Факультет Санкт-Петербургская школа гуманитарных наук и искусств

Егорова Ангелина Денисовна

Назаров Никита Сергеевич

THE IMPACT OF DIFFERENT TEXT PREPROCESSING PROCEDURES IN TEXT COMPLEXITY CHARACTERISTICS (ВЛИЯНИЕ РАЗЛИЧНЫХ ПРОЦЕДУР ПРЕДОБРАБОТКИ ТЕКСТА НА ЕГО ХАРАКТЕРИСТИКИ СЛОЖНОСТИ)

Выпускная квалификационная работа – МАГИСТЕРСКИЙ ПРОЕКТ
по направлению подготовки 45.04.03 Фундаментальная и прикладная лингвистика
образовательная программа «Языковые технологии в бизнесе и образовании»

Рецензент:
к. филол. н., старший
научный сотрудник
М. В. Худякова

_____

Научный руководитель:
к.ф.-м.н., доцент
Н. В. Станкевич

_____

Санкт-Петербург 2025

TABLE OF CONTENTS

INTRODUCTION 3
CHAPTER I. THEORETICAL FOUNDATIONS OF SENTIMENT ANALYSIS, STOCHASTICITY AND CHAOS 7
1.1 Concept of Sentiment analysis and its types 7
1.2 Sentiment analysis and opinion mining 11
1.3 Rule-based and Machine learning sentiment analysis 13
1.4 Sentiment Analysis' Features 15
1.4.1 Polarity 16
1.4.2 Subjectivity 18
1.5 Entropy and the concept of the Shannon's information theory 19
1.5.1 Entropy and complexity calculations 23
1.6 Complexity 25
1.6.1 Statistical complexity 25
1.6.1 Kolmogorov complexity 27
1.6.2 Lempel-Ziv Complexity 30
1.7 Lyapunov Exponent 32
Results and Conclusions of the First Chapter 33
CHAPTER II. SENTIMENT TRAJECTORIES 35
2.1 Choosing Literary texts for Analysis 35
2.2 Sentiment trajectories 36
2.3 Measuring Informativity in best-selling books 39
2.3.1 Shannon's entropy 40
2.3.2 Complexity 47
2.3.3 Lyapunov Exponent 47
2.4 Comparison of entropy-complexity measures of texts in different time periods 49
Results and Conclusions of the Second Chapter 51
CONCLUSION 52
REFERENCE LIST 54
APPENDIX A 62
APPENDIX B 68
APPENDIX C 69

INTRODUCTION

Language is an open system that is characterized by its flexibility and dynamism. In the modern world, there is an exponential growth in the volume of textual information, which necessitates the development of effective methods for its analysis. Nowadays there is a need in professional fields to determine the complexity of the text, for example in translation studies, teaching translation, marketing and educational spheres. For professional translators knowledge of text complexity contributes to understanding how much time they need to spend on translation, what types of reference materials they should use during the translation process, and assume the number and quality of mistakes allowed in this text. During translation training, knowledge of complexity indicators will make it possible for students to select texts for study that correspond to their real knowledge and skills. The informativity of a text can be assessed using various algorithms, including: the ratio of the number of terms to the total word count, the frequency of word repetition in previously translated texts, a readability index, and a purely mathematical calculation of information entropy.

In the present study probabilistic approach according to Shannon entropy and algorithmic approach of Kolmogorov will be considered.

To conduct an in-depth analysis of sentiment trajectories within the framework of entropy and time series, it is necessary first to analyze the tonality of words in the text.

Sentiment analysis was employed in work to generate time series representing the emotional content of the text. These time series will subsequently be used to quantify textual complexity and entropy. There are currently enough interdisciplinary studies on sentiment analysis and the concept of entropy, but few scientists have observed these concepts in one work. With the growing interest in NLP, the number of studies devoted to analyzing the randomness of time series based on textual data is also increasing. Sentiment

analysis, or text tonality analysis, in turn, is one of the most well–known processes for automatically determining the emotional coloring of a digital text. In this regard, we decided to combine two promising areas of research: computational linguistic analysis of emotions (or sentiment analysis) and mathematical measures of text complexity.

The **relevance** of the present study is that it is interdisciplinary: we touch upon the sentiment analysis that is related to the task of computational linguistics and information entropy, the matter of which relies in both branches of science: computational linguistics and theory of information.

The present work presents a mathematical **approach** to calculating entropy-complexity and uniform distribution.

The **main goal** of the study in the paper stands for the matter of how can Shannon entropy be applied in linguistics as well as in text analysis?

The other important questions that need to be solved are the following:

1. How does text preprocessing for further sentiment analysis impact entropy-complexity indicators?
2. How to prove that Shannon entropy unfolds stochasticity and chaos in the text?

In accordance with the objective, the following **objectives** were set:

- To do all pre-processing steps of texts using related Python Libraries: tokenization, stop words removal, PoS tagging, and lemmatization ;
- To present indicators of polarity and subjectivity and to perform it in a data frame;
- To transform words of a natural language into vectors;
- To present the sentiment trajectories of each literature masterpiece as one-dimensional time series on the entropy-complexity plane;
- To calculate entropy and complexity measures and place sentiment trajectories over entropy-complexity plane for corresponding time series

within the lower and upper theoretical boundaries and ascertain whether the texts are chaotic or not;

- To compare the trajectories of different texts to make conclusions about their chaoticity and informativity;
- Do the same tasks above without text preprocessing;
- To compare the results.

The **subject** of the study is a collection of 50 best-selling literary works published between 1974 and 2023, selected according to The New York Times Best Seller list, which is widely recognized as the leading ranking of popular books in the United States. The **research material** includes texts from various genres - such as adventure novels, thrillers, and dramas. They are chosen to enable a comparative analysis of how entropy-complexity indicators vary across different literary styles and historical periods.

The **object** of the study is sentiment analysis and entropy.

The **theoretical basis** for the work is formed by contemporary research and publications in the field of NLP, mathematics and computational linguistics.

The **novelty** of the study lies in the attempt to apply mathematical and physical theory to the field of linguistics using modern NLP and computational linguistics methods.

Practical significance: The results of this study can be applied in translation quality assessment by providing a quantitative method to evaluate the structural and emotional fidelity of translated texts.

Structure of the paper: Master's thesis consists of introduction, two chapters, conclusions on each chapter, conclusion, bibliography and appendices.

The **introduction** establishes the relevance and novelty of the work, the subject, the object, the goal and the objectives of the study and the approach of research.

The **first chapter** underlines the key concepts: sentiment analysis, its types, and features like polarity and subjectivity. It introduces Shannon entropy,

Kolmogorov complexity, and the Lyapunov exponent as tools to measure linguistic chaos and structure. These concepts form the basis for analyzing sentiment dynamics in texts.

In the **second chapter** theory applies to practice. After using 50 best-selling books, it constructs sentiment trajectories and computes entropy, complexity, and Lyapunov values, comparing results with and without preprocessing. The results show that preprocessing affects the detection of chaotic patterns and informativity in texts.

The **conclusion** describes the results of the work: the study shows that preprocessing reduces noise and highlights deeper patterns in sentiment analysis, the effectiveness of the methods used is analyzed. Finally, the possible directions for the further development of this project are proposed.

Egorova A. performed text preprocessing, conducted a literature review on chaos, stochasticity, entropy, and complexity, constructed the sentiment trajectories, in addition to drafting the theoretical framework. Nazarov N. carried out text analysis with focus on statistical complexity and sentiment analysis, and contributed to writing the overall theoretical sections, including the choice of literary texts.

The intermediate results of research have been tested in the scientific journal “Culture and text” and were presented in the international scientific and practical conference “Natural Language Processing (NLP) for Linguistics, IT and Business”.

Approbation of the work:

1.Egorova A., Stankevich N. (2024). Sentiment trajectories like a tool for text analysis, Culture and text, 2(57), 169-180.

2. Egorova A., Nazarov N. (In press, 2025) The Impact of different text preprocessing procedures in text’s sentiment trajectories, International scientific and practical conference “Natural Language Processing (NLP) for Linguistics, IT and Business” [Tomsk, Russia].

CHAPTER I. THEORETICAL FOUNDATIONS OF SENTIMENT ANALYSIS, STOCHASTICITY AND CHAOS

Our present work delves into the employed techniques, underscores the importance of information entropy in NLP and examines the evolving spread of emotion analysis for the last decade.

1.1 Concept of Sentiment analysis and its types

Sentiment analysis is a field of natural language processing (NLP) that focuses on identifying and extracting subjective information from text. The primary goal is to determine the attitude, emotions, or sentiments expressed by a writer or speaker regarding a particular topic, product, service, or event. This sentiment is often classified as positive, negative, or neutral, though more nuanced analyses can detect specific emotions like joy, anger, sadness, or surprise. One of the important aspects of text analysis is the identification of sentiment (emotional tone), which finds application in a wide variety of fields, from monitoring public opinion to analyzing consumer behavior. The existing methods of sentiment analysis, as a rule, are focused on determining the general text tonality, while the dynamics of sentiment change (sentiment trajectories) remains insufficiently studied, despite its potential informative value.

One of the first works in which the term sentiment analysis is used is the work of T. Nasukawa and J. Yi. The authors applied sentiment analysis to words from news articles after doing all pre-processing steps, including POS tagging (Nasukawa & Yi, 2003, p. 72–73). It is interesting to note that authors use such a method of analyzing sentences as *shallow parsing*. The method consists in the initial identification of separate units of sentences (nouns, verbs), and then the phrasal groups they are part of. Researchers proposed to single out positive, negative and neutral polarity. After obtaining the results of the shallow parser, authors analyze the syntactic dependencies among the phrases and look for

phrases with a sentiment term that modifies or is modified by a subject term. However, this approach did not work in examples with ambiguity in complex sentences, which relates to“the presence of two or more possible meanings within a single word”, periphrasis, irony (“she is grammyless”). The reason for this is the fact that domains and data types are expanded, and the precision of some difficult data may go down to about 75 %. As the prerequisite of study, researchers need to modify the current system, adding new sentiment terms for new domains. Moreover, the authors are going to resolve the issue with various discourse processing methods including anaphora resolution in the future works (Nasukawa & Yi, 2003, p. 75-76).

Liu (2012) expressed support for the idea that sentiment analysis has become one of the most popular research fields in natural language processing (NLP). He made an extensive summary, emphasizing his goal in presenting the applications and development of sentiment analysis in the business and social sphere. The author sees his goal as very helpful in the area of qualitative and quantitative analysis of opinions.

As sentiment analysis has been the subject of computational linguistics for a long time, the authors describe steps in carrying out the sentiment analysis. Firstly, they have done pre-processing of natural language texts, using related programming libraries, such as spacy, natasha and others. He mentions that in sentiment analysis “each word is mapped into a float number, ranging from -1 to 1”, and “The value -1 corresponds to words that evoke the most negative emotions; the value 1, to words that evoke the most positive emotions”. Emotional analysis underlies the work on experience clues by Reagan. et al. (2016), in which he investigates six core emotional arcs within the storyline. The similar results may be found in works of Min and Park, who propose to explore a literary masterpiece by calculating the mean Sentiment Polarity Index for each chapter to ascertain a chapter “topic” (Min & Park, 2016).

Mercha and Benbrahim (2023) emphasize the significance of opinionated textual data in the social world. Researchers prove that “the majority of existing sentiment analysis surveys focus on monolingual methodologies, with English being the most extensively researched language” (Mercha & Berbrahim, 2023). They have carried out multilateral research in the frame of state-of-art methods for sentiment analysis across multilingual approaches. The authors also discuss the methods of machine learning and deep learning models, providing new research directions.

Several questions regarding sub-tasks of aspect-based sentiment analysis (ABSA) remain to be addressed. Researchers Shuang, Gu, Li et al. (2021) bring some information about the background of the problem. Actually, in recent years, a lot of neural models have performed greatly in aspect-level sentiment-classification tasks. However, as emphasized by the study's authors, their prediction accuracy is not always reliable, because of the noises in some biased training samples. Therefore, other studies scarcely applied part-of-speech tagging. Shuang et al. concur that the part-of-speech information becomes an integrated part of the research, as it helps researchers to analyze a structure and coherence of the sentence. Moreover, as has been previously reported in other literature, certain parts- of- speech (POS) categories would be the strong indicators of the sentiments (Shuang et al., 2021). Researchers prove that only three kinds of POS contribute to sentiment and subjectivity: adjectives, verbs and adverbs. To concentrate on these three specific POS categories, Shuang et al. propose the interactive POS-aware network (IPAN), which includes components that are designed for differentiating adjectives, adverbs and verbs from other POS categories. Moreover, Shuang et al. developed an upgraded target-context gate called gated ReLU Units (GReU). As a result, the authors designed an innovative POS-highlighting attention mechanism that focuses only on specific parts of speech. Authors managed to show the success of their model comparing the novel gating units with the existing one. Comparing the full

model IPAN with its three ablated models, authors concluded that removing any of the three components deteriorates the end results. The considerable advantage of the work lies in explicit visualization of the POS-highlighting mechanism.

They prove the positive results from applying these methods on model performance. Therefore, to conduct fundamental analysis containing syntax information Multiple Syntactic Structure (MSS) is proposed. Their experiment was based on three datasets in which such sub-tasks as Aspect Sentiment Triplet Extraction (ASTE), Aspect Term Extraction (ATE), Aspect-oriented Opinion Extraction (AOE) and Opinion Term Extraction (OTE) are considered. The research materials were most widely used datasets for the aspect-level sentiment-classification task: Laptop, Restaurant and Twitter datasets. The goal of the experiment was to identify the aspect polarity of a sentence with the corresponding target in the most precise manner. The proposed approach is compared against baseline models, which only focus on a single or subset of sub-tasks in a pipeline. The authors conclude that a language model called BERT is used to obtain semantic embeddings, and can further improve the performance of ABSA subtasks. Research showed the following results:

- development an interactive POS-aware network (IPAN) as a tool for detection POS information;
- designing a POS-filter gate to generate transformed word embeddings;
- designing POS-highlighting attention mechanism with a limiting condition to assigning high attention scores for POS categories;
- development of a target-context gate called gated ReLu Units (GreU).

Authors emphasize the aspect (2021) that is often overlooked in sentiment analysis: categorization by parts of speech. It is regarded as a significant step for the successful operation of the model. Their approach is completely different from other approaches, as it is built on the idea of different contributions to the sentiment polarity depending on multiple POS categories. To conclude, Shuang et al. designed a state-of-art POS-highlighting attention mechanism to prove that

the model fully focused on adjectives, adverbs, and verbs. However, as the authors note earlier, more work is necessary to transfer the learned knowledge from one domain to other domain datasets with the help of transfer learning.

As it has been noted earlier, more work is necessary to investigate whole texts within semantic and sentiment trajectories.

1.2 Sentiment analysis and opinion mining

There is also a term that is often identified with the above mentioned such as *opinion mining,* which is a feature of sentiment analysis, as polarity and subjectivity. Sentiment analysis is a generic concept to which the term opinion mining refers. It is also known as *aspect-based sentiment analysis* in Natural Language Processing (NLP). The term is defined as the process of identifying sentiment aspects in a text by focusing on specific aspects, sentiment polarity, and opinion terms related to the text (Ezzameli & Mahersia, 2023). So, sentiment analysis mainly studies opinions which express or imply positive or negative sentiments (Liu, 2012). It is also known as aspect-based sentiment analysis in Natural Language Processing (NLP). One of the first and most famous works in opinion mining was presented in 2002 by Pang and Lee (Pang, Lee, & Vaithyanathan, 2002). They used three machine learning algorithms of NaiveBayes, MaximumEntropy and support vector machine to classify film opinions into two groups of positive and negative (Amir & Alireza, 2017).

The opinions can be classified according to their meaning and pragmatic functions. The first type of opinion that we will discuss is *regular opinion*, another type is called *comparative opinion*. A regular opinion is often referred to simply as opinion in the literature and it has two main subtypes: *direct opinion* and *indirect opinion* (Liu, 2006, 2011, 2012). A *direct opinion* refers to an opinion expressed directly on an entity or an entity aspect, whereas an *indirect opinion* is an opinion that is expressed indirectly on an entity or aspect

of an entity based on its effects on some other entities. For example, the sentence “This camera is great” describes the opinion expressed directly on the object aspect (camera). Indirect opinion is more difficult to recognise than *direct opinion*. One needs to know whether some desirable and undesirable state is before or after some action (Liu, 2006), e.g., the phrase “After taking sleeping pills I became addicted to it” expresses a negative opinion on the pills, but the sentence “Since I have had sleep disorders, I started taking sleeping pills” does not express a sentiment or opinion on the pills because “sleep disorders” (which is negative) happened before taking medicine.

A comparative opinion is a type of opinion which is expressed by comparing multiple targets on a shared aspect (Wang et al., 2022), e.g., “This picture is more colourful than the previous one”. Comparative and superlative adjectives are frequently used in such sentences.

There is also another classification of opinions that is based on expressiveness: explicit and implicit opinions. An explicit opinion is a subjective statement that gives a regular or comparative opinion, e.g.,“This forest is wonderful,” and “This forest is more wonderful than where we have been.”

An implicit opinion is a statement that only implies a regular or comparative opinion, e.g., “I bought a cake yesterday, and it was too sweet”, and“ This MacBook works faster than previous.”

Unfortunately, less work has been done on implicit opinions (Zhang & Liu, 2011; Liu, 2012).

Sentiment analysis can be defined as a natural language processing (NLP) technique which defines positive or negative evaluation expressed through language. This method has gained great importance in various popular fields, besides linguistic and computer sciences, for example, business, marketing, customer service, social media analysis, etc.

Overall, sentiment analysis is one of the most widely known natural language processing (NLP) tasks. The analysis of emotions consists in determining the emotional meaning of an expression in natural language.

1.3 Rule-based and Machine learning sentiment analysis

It's generally known that sentiment analysis generally falls into three categories:

- Rule-based algorithms
- Machine Learning models
- Lexicon-based systems

The three major approaches to NER are based on lexicon, rules, and machine learning (Gudivada, Rao, & Raghavan, 2015). Rule-based approach is a practical approach to analyzing text without training or using machine learning models. This approach results in rules based on which the text is labeled as positive, negative or neutral. These rules are also known as lexicons. Hence, the Rule-based approach is called the Lexicon-based approach. These include *TextBlob, VADER, and SentiWordNet.* A sentiment analysis dictionary contains information about the emotions or polarity expressed by words, phrases, or concepts (Amirkumar, Orynbekova, Talasbek, Ayazbayev & Cankurt, 2024). This approach however has several limitations. Firstly, the dictionary-based method is not able to handle ambiguity. Secondly, it does not work with different dialects and informal or slang words, as well as acronyms and shorthand. Besides this, when units transfer to other languages, they suffer from translation problems and cultural differences (Kaity & Balakrishnan, 2020).

With a machine learning (ML) approach, an algorithm is used to train software to estimate sentiment level in a block of text using words that appear in the text as well as the order in which they appear. Engineers and researchers use sentiment analysis to teach their software how to identify emotion in text,

similarly to how we, humans, do. ML models continue to “learn” from the data they are given, hence the name “machine learning” (Amirkumar et al., 2024).

Here are a few of the most commonly used classification algorithms:

- Linear regression - calculates how the X input (words and phrases), relates to the Y output (polarity). LR will determine where words and phrases fall on a polarity scale from positive to negative and everywhere in between (Tang et al., 2022).
- Naive Bayes - algorithm that uses Bayes’ theorem to categorize words in a batch of text. Bayes classifier learns the probability of each word appearing in positive, negative, or neutral text from the training data (Wongkar & Angdresey, 2019)
- K-Nearest Neighbor (KNN) - classification system, which is based on the concept that similar data points stay close to each other. This model helps to determine what words and phrases are positive, negative, or neutral based on the data that exists around them.
- Support vector machines - quick and efficient classification algorithm used to solve two-group classification problems.
- Random Forest is an ensemble learning technique in which many decision trees are trained and where each tree carries a data-supplied sample of observations.
- Deep learning (DL) - also known as an artificial neural network, which is considered to be an advanced machine learning technique that links together multiple algorithms to mimic human brain function. Models related to Deep learning include models such as BERT, RoBERTa and other Transformer-based architectures, that are used for text sentiment analysis (Umarani, Julian, & Deepa, 2021).

We should point out the pros and cons of both approaches. The main advantages of rule-based approach are: accuracy, ease of use and swiftness. But there are a few limitations: limited scope and restricted intelligence.

Machine learning systems provide the following benefits to organizations: adaptability, self-learning and scalability. On the other hand, they also require greater technical expertise from teams to develop effective applications. Moreover, building a sufficiently large and relevant data set can be a resource-intensive and time-consuming process (Umarani et al., 2021).

One must admit that rule-based sentiment analysis is one of the very basic approaches to calculating text sentiments. It only requires minimal pre-work and the idea is quite simple, this method does not use any machine learning to figure out the text sentiment. On the contrary, instead of being explicitly programmed, machine learning models, a subset of AI, learn from data to enhance their performance in specific tasks. Rule-based systems thereby rely on predefined dictionaries, while AI-powered machine learning learns continuously from data.

1.4 Sentiment Analysis' Features

Sentiment analysis is used in many professional spheres: social media, marketing, medicine, literature. Berardinelli, Gaber and Haig (2013) highlighted that analysing students' feedback using sentiment analysis techniques can identify the students' positive or negative feelings, or even more refined emotions, that students have towards the current teaching. The authors represented a table where the emotion types were categorised into Love, Joy, Anger, Frustration, and Neutral. The student can express his feelings in short expressions or words (Ezzameli & Mahersia, 2023; Geetha et al., 2024). Tian Feng et al. (2009) created patterns to find what words are associated more with emotions, they also give solutions to relieve the e-learner, for example, when he feels depressed as his teacher does not like his essay (Feng et al., 2009).

Sentiment analysis comes in many forms - depending on the tasks and objectives you pursue. Linguistic sentiment is a measurement of subjectivity as

either a positive or negative opinion in text. In the Chinese blogs, for example, the positive, negative and objective emotion scores are added in the model and show their significance to find influential users (Shuang et al., 2021). Traditionally there are two types of sentiment analysis, according to its further usage: *fine-grained sentiment analysis* and *aspect-based sentiment analysis* (Yan et al., 2022). As a measurement of opinions and affective states, a sentiment score generally consists of two aspects: polarity and intensity (Tian, Lai & Moore, 2018). A key aspect of sentiment analysis is polarity classification. This polarity can be expressed as a numerical rating known as a sentiment score. Fine-grained, or graded, sentiment analysis is a type of sentiment analysis that groups text into different emotions and the level of emotion being expressed (Jones, 2024). Categories can expand beyond just positive, neutral and negative. For example, one may choose to use five categories, e.g., negative, neutral, positive and very positive. In its most complete iteration, fine-grained analysis is conducted to define both the sentiment target (topic) and its intensity at a phrase or clause level (Wida & Setiawan, 2023).

1.4.1 Polarity

A basic task in sentiment analysis is classifying the polarity of a given text at the document, sentence, or feature/aspect level - whether the expressed opinion in a document, a sentence or an entity feature/aspect is positive, negative, or neutral. Advanced, “beyond polarity” sentiment classification looks, for instance, at emotional states such as enjoyment, anger, disgust, sadness, fear, and surprise (Vong et al., 2019). The parameter can be in the range from -100 to 100, but most often the sentiment score is between -1 and 1, indicating 0-neutral sentiment.

These are the main components of the concept of polarity:

- Polarity Classification: This involves assigning a sentiment label (positive, negative, neutral) to a given text. For instance, the sentence “I love this product” has a positive polarity, while “I regret buying this item” has a negative polarity.
- Lexicon-Based Approaches: These methods utilize predefined dictionaries where words are associated with specific sentiment scores or polarities. The overall sentiment of a text is determined by aggregating the polarities of individual words.
- Machine Learning Approaches: These involve training models on labeled datasets to learn patterns associated with different sentiment polarities. Such models can capture more nuanced expressions of sentiment compared to lexicon-based methods (Liu, 2012; Zhang, Wang & Liu , 2018).

In NLP, polarity is predominantly associated with sentiment analysis, a subfield focused on determining the emotional tone of text. Sentiment polarity indicates whether a piece of text expresses a positive, negative, or neutral sentiment, forming the basis for opinion mining and affective computing. This application is particularly relevant in analyzing user-generated content, such as product reviews, social media posts, and developer communications.

A notable study, “Polarity Calibration for Opinion Summarization” (Polarity Calibration for Opinion Summarization), addresses the challenge of ensuring that automated summaries reflect the sentiment distribution of input texts. The paper introduces polarity calibration, a method to align summary polarity with input, mitigating biases where models amplify majority opinions and neglect minorities (Lei et al., 2024)

1.4.2 Subjectivity

Textual information in the world can be broadly categorized into two main types: facts and opinions.Facts are objective expressions about entities, events and their properties. Opinions are usually subjective expressions that describe people's sentiments, appraisals or feelings toward entities, events and their properties (Bing, 2010). Subjectivity as a polarity is also scored, usually from -1 to 1.

However, subjectivity is not a simple binary attribute but rather exists on a spectrum. It is often signaled through a variety of linguistic markers, including evaluative adjectives , adverbs and verbs. Modal verbs (e.g., "might", "could", "should") and hedges (e.g., "sort of", "kind of", "apparently") can also indicate subjectivity by expressing uncertainty or limited commitment to a statement. The frequent use of personal pronouns (e.g., "I", "me",' "my") and subjective intensifiers (e.g., "very", "really", "extremely") further contributes to a more subjective tone. Furthermore, the concept of subjectification (Traugott & Dasher, 2002) highlights how meanings evolve over time, with objective expressions often acquiring more subjective interpretations through processes of grammaticalization. The notion of stance (Du Bois, 2007), encompassing the speaker's attitude, feelings, and judgments, is also central to understanding subjectivity. The accurate identification of subjectivity is crucial for various applications in Natural Language Processing (NLP). It is a foundational step in sentiment analysis (Pang & Lee, 2008), where the goal is to determine the overall sentiment expressed in a piece of text. Subjectivity detection is also vital for opinion mining (Liu, 2012), allowing for the extraction of subjective opinions from large datasets. Moreover, it plays a key role in fact-checking and bias detection (Rashkin et al., 2017), helping to identify biased or misleading information in news articles and social media. As such, a robust understanding of subjectivity and its linguistic manifestations is essential for advancing

various NLP tasks and promoting more informed and nuanced analyses of textual data.

1.5 Entropy and the concept of the Shannon's information theory

Entropy arises exclusively in dissipative systems, like language. The concept of entropy was introduced into scientific circulation in 1865 by R. Clausius to characterize the processes of energy conversion (Chumak, 2011). The term entropy is also used in classical mechanics as a "characteristic of dynamic chaos in systems with instability of movement". In theory of information entropy means "a measure of the uncertainty of a certain situation" (Korotaev, 2013). The entropy of a language is a statistical function of a text in a particular language, or of the language itself, which determines the amount of information per unit of text (Schneier, 2018). In modern linguistics, this concept is significant and capable of replacing the term information (Nekipelova, 2013, p.3). Thus, information entropy determines the measure of stochasticity and randomness of the text.

The concept of entropy provides deep insight into the direction of spontaneous change for many everyday phenomena. Its introduction by the German physicist Rudolf Clausius in 1850 is a highlight of 19th-century physics (Drake, 2025). Research on stochasticity and time series has increased with growing interest in NLP. Semantic trajectories of natural texts require competent application of the right approach. For example, researcher V. Gromov (2023) devotes his works to sentimental analysis of literary texts and notes the main tendency in entropy and complexity values in data frames. In addition, he tried to determine the sentiment and semantic trajectory of a literature masterpiece using mathematical methods such as Shannon Entropy and the largest Lyapunov exponent (2015). The object of his research is Russian and English outstanding literature.

Language is a kind of programming code that allows us to express our thoughts and feelings. When a person speaks or writes, he is a source of information. The Shannon entropy estimates the average amount of information (mathematical expectation) contained in the values of a random variable. According to Shannon's formula, the information added by each new character is the logarithm of the probability of its occurrence, the basis of which is the number of characters in the alphabet (Gromov, 2023). With the help of information theory, it is possible to reduce the number of letters in words without loss of meaning. Then the brain, based on a limited amount of information from the “truncated” text, will itself complete the missing elements to restore meaning. It turns out that all our languages are redundant to one degree or another, i.e. they contain sounds or letters that are not strictly necessary to convey a message. Claude Shannon estimated the redundancy of the English language at 75% information does not exist without a physical medium; it is discrete and measured in bits. Information in the narrow sense is Shannon entropy, that is, the degree of uncertainty and novelty in a message or the average amount of information per message character (Chumak, 2011). Therefore, the more uncertain the data, the more information is required to describe it.

The definition of information entropy was introduced in 1948 by Claude Shannon. In information theory it is the average rate at which information is produced by a stochastic source of data (Nasukawa & Yi, 2003). It quantifies the amount of information in the variable. This term was adapted for different processes and can be used for characterization of chaoticity, or stochasticity of the processes. The approach implies that, for a given time series, a user calculates two characteristics - entropy and complexity and then compares the position of this point with the lower and upper theoretical boundaries to specify time-series type. The main advantage of this algorithm is that it is possible to work with only one-dimensional time series.

Entropy as the degree of dispersion of meaning can be defined as follows: is a measure of the uncertainty or variability of information, which is often used to describe the degree of dispersion or dispersion of meaning in a message or text. The higher the entropy, the more uncertainty or ambiguity is contained in the message, which makes it less predictable or unambiguous. (Pavlovskaya & Illina, 2023, p. 80).

The use of entropy in linguistics allows for a quantitative analysis of the chaoticity coefficient of texts, which opens up new opportunities for studying language and its structures (Gromov, 2023).

According to Gromov and Migrina (2017), language fractal structures can be identified as the sets of structure in natural language that appear in the same cluster, i.e. in the similar languages. The authors reveal the idea that "the texts of a language are "avalanches" flowing down its word co-occurrence graph as complex networks and random walks. The goal of the article is to analyze language stochasticity, exploring "real-world semantic "avalanches" generated by a particular language". The choice of the research material is due to the fact that both Russian and English language are rich of diverse corpora of texts, moreover they have qualitative difference in grammar structure: Russian is an inflected language, while in English inflections are rather rare; Russian is characterized by flexible word order in a sentence, whereas word order in English language is strict, and rare exceptions are constrained by stringent rules.

It's important to pinpoint the fact that authors used two different methods to test the statistical hypothesis. The first one is a technique of data collapse, the second one is based on the Kolmogorov-Smirnov (KS) criterion. Both of these methods not only evaluate the exponents but also remove the tiniest parts of the sample that usually do not fit a power-law. The first method also cuts off the biggest non-fitting elements. It is worth mentioning that this phenomenon seems to be a vital characteristic of the real world information following power-law distributions.

In (Gromov et al., 2023) authors calculated different measures of sentiment and semantic trajectories complexity. They've shown that such trajectories for different languages are chaotic, and indicators of chaoticity and complexity, like entropy, or the largest Lyapunov exponents, can be used for estimation of translation quality of literature masterpieces. Nevertheless, it remains unknown how the sentimental analysis of time series is carried out specifically for parts of speech.

As for the results, Gromov and Dang calculated minimum, average, maximum entropies and complexities for every literature masterpiece. They also found mean entropy and complexity deviation from original. Authors analyze Russian-English translation for novels "Crime and Punishment" by F. M. Dostoevsky and "Anna Karenina" by L. Tolstoy. Translations which were considered, were made by outstanding translators as K. Garnett by F. Dostoevsky and "Anna Karenina" by L. Tolstoy. For English-to-Russian translation authors consider several works by translators from the 20-21st centuries: A. Gorkovenko, E. Lann and A. Krivtsova and V. Lukianskaya. As for the English-to-Russian translations, authors also calculate the entropy and complexity of a machine word-by-word translation and a machine translation performed by a Facebook translator. The results turn to be quite obvious: the translations closest to original were made by human, namely professional in her field, K. Garnett (1.49% entropy deviation and 0.99% complexity deviation in the translation of novel "Anna Karenina" for n=3, d=5). Machine translation shows worse results: -0.25 for entropy and 10.96% for complexity for the same novel and indicators. It is natural that word-by-word translation appears to be the farthest from the indicators of entropy and complexity in original masterpieces (-3.29% for entropy and 1.90% for complexity for the same data).

The empirical probability density function showed good results, the authors discovered the problem of diachronic unity, but existing studies have

failed to address that they limited themselves to the corpus created by the 20th century for both languages.

1.5.1 Entropy and complexity calculations

Gromov and Dang also managed to represent the cluster of different languages on the entropy-complexity plane. Drawing to the author's conclusion, results based on the entropy-complexity plane do not corroborate its claim to be equidistant from all languages. Also, Gromov proved that poetry is closest to the traditional law, and therefore poetry can be considered as a component of the basis of language. However, it seems to be equidistant from all Indo-European languages. It is proposed that in future studies it would be fruitful to explore other languages in the framework of language typology.

The calculation of the complexity measure was used in the work, and its calculation was based on the formulae where Q is a Jensen–Shannon divergence between the given and the uniform distributions (P, P_e) multiplied by conventional Shannon entropy, normalized to its value:

$$C[P] = Q[P, P_e] * H[P].$$

The lower and upper theoretical boundaries were placed on the entropy-complexity plane and its position was compared with the position of given time series (sentiment trajectories). Simple deterministic processes occupy the bottom left corner of the plane, stochastic processes cover the bottom right corner. Chaotic processes occupy the areas near the upper curve (Fig.1).

The theoretical boundaries were calculated as the conventional measure of complexity:

$$C = H * D.$$

Parameter *H,* which is identified with the thermodynamic entropy, stands for information or entropy, and parameter D for disequilibrium, respectively (Lopez-Ruiz et al., 2010).

When an isolated system valves with time, the complexity cannot have any possible value in a C versus H, but it must stay within certain bounds, *Cmax* and *Cmin*. These are the maximum and minimum value of C for a given H (Lopez-Ruiz et al., 2010). The largest and smallest values of the function (*fmin, fmax)* were obtained using the Sympy library.

The lower and upper theoretical boundaries were placed on the entropy-complexity plane and its position were compared with the position of given time series (sentiment trajectories). Simple deterministic processes occupy the bottom left corner of the plane, stochastic processes cover the bottom right corner, whereas chaotic processes occupy the areas near the upper curve (Figure 1).

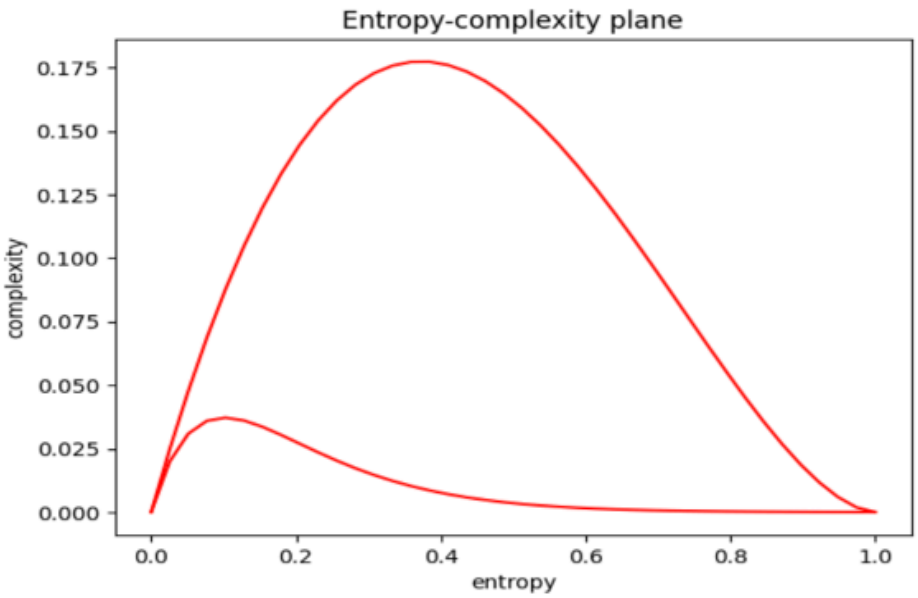

Figure 1. Theoretical boundaries on entropy-complexity plane

The low boundary was calculated as minimum envelope complexity that traverses all the discontinuous derivative points in the *Dmin* versus *H.* In total, parameters were obtained: *N* and *n*. *N* is the total number of accessible states, and *n* is the number of states where *f_i* = 0. *N* is the total number of words. This is shown in Figure 1.

After the theoretical boundaries were drawn, the sentiment trajectories of each text, represented as one-dimensional time series, were placed on the entropy-complexity plane. Time series can be defined as follows: a sequence of observations of a certain parameter at different points in time.

For a relatively small time series, the probabilities are close to zero, the distribution of probabilities is close to uniform.

The entropy of the text reflects the degree of chaos or uncertainty of the probabilities (events) in the text. The higher the entropy, the more diverse ideas are used in the text, and the more "complex" or "unexpected" this text is in terms of predicting the next event.

1.6 Complexity

Complexity characterizes the behavior of a system or model whose components interact in multiple ways and follow its local rules, leading in particular to non-linear system, randomness, collective dynamics, hierarchy (Johnson, 2001). This is where the term "complex system" originally comes from. A complex system is a system consisting of many interacting components (also known as subsystems), as a result of which it gets new properties (Bar -Yam, 2002).

1.6.1 Statistical Complexity

Statistical Complexity is a measure used to describe how structured or organized a system is, while also accounting for randomness. Unlike measures that only focus on how unpredictable a system is (such as entropy), statistical complexity tries to capture both the amount of information and the degree of structure present in the data (Crutchfield & Young, 1989). Generally, applications cover quantum systems, software-architecture networks (modeling components and interconnections via state-probabilities), and even edge-map scoring in image processing. In text complexity analysis, SC can reveal how

syntactic or lexical distributions balance predictability and novelty, offering insights that pure entropy measures miss.

A complementary notion arises in computational mechanics, where Statistical Complexity is the Shannon entropy of a system's causal states - the minimal set of historical patterns needed to optimally predict the future. First formalized by Crutchfield & Young, SC quantifies memory stored in the process's ε-machine (a model that helps us understand and predict patterns in a data sequence by identifying the minimal amount of memory needed to make good predictions). Unlike SC's focus on probability-distance products, SC directly measures the hidden–Markov-model complexity underlying temporal sequences.

In text analysis, SC can be used to assess how much contextual history (e.g., previous words or syntactic frames) is needed to predict upcoming words, reflecting deep structural complexity beyond surface-level probabilities.

Kolmogorov Complexity of a string x (which we will touch on in more detail later) has a different meaning and is defined as the length of the shortest program on a universal Turing machine that outputs x. (Shen, 2024) It is a fundamental notion of algorithmic randomness. However, K(x) is uncomputable in general, and practical estimators rely on bounds or approximations. Moreover, K treats all strings uniformly, ignoring any probabilistic model over ensembles. In contrast, Statistical Complexity measures are computable given empirical distributions or inferred models. They explicitly leverage Shannon entropy and distributional structure (e.g., disequilibrium or causal-state probabilities), making them more readily applicable to real-world data sets.

Lempel–Ziv Complexity (LZC) approximates algorithmic complexity by counting the number of distinct substrings ("phrases") as a sequence is scanned left to right. Each time a new substring appears, the LZC counter increments, offering a fully computable estimate of string compressibility. While LZC

effectively measures repetitions and compressibility, it does not directly incorporate a notion of disequilibrium or the entropy of an underlying distribution. Statistical Complexity, on the contrary, balances entropy with structural measures (e.g: Jensen–Shannon divergence in generalized SCMs), capturing both randomness and organization. Moreover, the computational-mechanics variant SC can detect long-range dependencies and causal architectures that single-pass substring counts may miss.

By selecting the appropriate Statistical Complexity formulation - whether López-Ruiz's entropy - disequilibrium product or Crutchfield's causal-state entropy - researchers can tailor complexity analyses to emphasize distributional balance, predictive memory, or both. This makes SC a versatile tool for probing the complicated structure of order and randomness in textual data, complementing, but also fundamentally differing from algorithmic and compression-based complexity measures.

1.6.2 Kolmogorov complexity

In theoretical machine learning, statistical complexity is a notion that measures the richness of a hypothesis space. At its core, Kolmogorov complexity is a way of measuring the amount of "information" in a piece of data - typically a string of text or bits. In simple terms, the Kolmogorov complexity of an object is defined as the length of the shortest computer program (written in a fixed, universal programming language) that can generate that object and then stop. (Li & Vitaniy, 1993)

This concept is important for various reasons:

- It measures Inherent Complexity:
  Unlike other measures that depend on probability distributions (like Shannon entropy), Kolmogorov complexity focuses on the inherent structure of the data. For example, a string that repeats a simple pattern

(like “abababab…”) can be generated by a very short program (“print ‘ab’ 100 times”), meaning it has low complexity. In contrast, a string that appears random (with no discernible pattern) will require a program nearly as long as the string itself.

- Helps us to understanding randomness:
  In this framework, randomness is understood as the absence of any shorter description. If a string’s shortest description is nearly the same length as the string, it is considered algorithmically random or incompressible. This concept helps us rigorously define what it means for data to be “random” or “complex.”
- It is universal
  Although the exact value of Kolmogorov complexity can change slightly depending on the programming language or Turing machine used, the differences are bounded by a constant. This means the measure is universal enough to compare the complexity of different objects in a meaningful way.

- Theoretical and Practical Impact:
  While Kolmogorov complexity itself is not computable (there is no general algorithm that, given any string, will produce its exact complexity), it underpins many theoretical results in computer science and mathematics. It also inspires practical methods in data compression, machine learning, and statistical inference (for instance, through the Minimum Description Length principle) (Li & Vitányi, An Introduction to Kolmogorov Complexity and Its Applications, 2008).

In NLP, researchers use the idea of complexity to compare different languages, texts, or even different writing styles. For instance, by estimating the complexity of sentences or entire documents, one can gauge their predictability.

Texts that are more predictable (e.g., formulaic news reports) tend to be more compressible and thus have lower Kolmogorov complexity than more creative, less predictable texts like poetry.

Kolmogorov complexity also inspires methods for clustering and measuring the similarity between texts. One such method is the Normalized Compression Distance (NCD). NCD approximates the similarity between two texts by comparing how well they can be compressed together versus separately. This technique has been applied to tasks like document clustering, language classification, and even authorship attribution.

A key paper on this approach is Cilibrasi & Vitányi (2007), "The Google Similarity Distance" which explores compression-based similarity measures.

The concept of entropy, in the context of information theory, generally refers to the unpredictability or randomness of a source. When we talk about complexity entropy in relation to Kolmogorov complexity, we are interested in how the complexity of a text (or any data) changes as its length increases, or how "surprising" the next piece of information is given what we have seen so far.

In many applications (especially in NLP) - entropy is used to measure the average unpredictability of a text. A higher entropy value indicates that the text is less predictable, which usually corresponds to a higher Kolmogorov complexity. Conversely, texts with lower entropy are more predictable and have lower complexity.

By analyzing the entropy of text, researchers can make decisions about language models, such as determining the best way to predict the next word in a sentence. This has direct implications in applications like speech recognition, machine translation, and text generation. The interplay between entropy (as a measure of average uncertainty) and Kolmogorov complexity (as a measure of the minimal description length) provides a robust theoretical framework for understanding and improving these models. (Vitányi & Li, 2000).

1.6.2 Lempel-Ziv Complexity

Lempel-Ziv complexity (LZ complexity) is a fundamental measure in information theory, quantifying the complexity of finite sequences based on their compressibility. Introduced by Abraham Lempel and Jacob Ziv in 1976, it serves as a practical approximation to the theoretical concept of Kolmogorov complexity. This paper provides an in-depth examination of LZ complexity, its computation, applications, and its relationship with Kolmogorov complexity. The concept of sequence complexity is pivotal in various domains, including data compression, randomness testing, and information theory. While Kolmogorov complexity offers a theoretical foundation, its uncomputability necessitates practical alternatives. LZ complexity emerges as a computable measure, reflecting the repetitiveness and structure within sequences.

LZ complexity quantifies the number of distinct substrings encountered as a sequence is parsed from left to right, where each new substring represents a novel pattern not previously observed. Formally, for a sequence S of length n, the LZ complexity C(S) is the count of such unique substrings identified during this parsing process. An efficient algorithm exists for computing LZ complexity in linear time - the algorithm involves iteratively scanning the sequence to identify the longest substring that has not been previously encountered, incrementing the complexity count each time a new substring is found.

LZ complexity has been applied in various fields:

- Data Compression: it creates the basis for algorithms like LZ77 and LZ78, that are foundational in such vastly known compression formats such as ZIP and GIF.
- Randomness Testing: Used to assess the randomness of sequences, especially in cryptography.
- Neuroscience: Applied in the analysis of EEG (electroencephalography) data to detect complexity in brain signals.

Now, speaking of similarities:

- Their goal is to measure the complexity or randomness of sequences.
- Lower values in both indicate that there is more regularity or compressibility.

And differences:

- LZ complexity is computable in linear time, while Kolmogorov's is uncomputable.
- Kolmogorov complexity depends on the choice of universal Turing machine, whereas LZ complexity is model-independent.
- Granularity: Kolmogorov complexity provides a theoretical lower bound on compressibility, meanwhile LZ complexity offers users a practical estimate.

Due to its computability, LZ complexity is widely used in practical applications where an estimation of sequence complexity is required. Kolmogorov complexity, while theoretically profound, serves more as a conceptual framework for understanding the limits of data compression and randomness.

In conclusion, Lempel-Ziv complexity serves as a practical and computable measure of sequence complexity, closely related to the theoretical construct of Kolmogorov complexity. While Kolmogorov complexity provides a foundational understanding of information content, LZ complexity enables real-world applications across various domains. Understanding both measures helps us understand the concept of data complexity and the principles underlying data compression and randomness.

1.7 Lyapunov Exponent

A Lyapunov exponent quantifies the rate at which nearby trajectories in a dynamical system either converge or diverge over time. Specifically, it measures how small differences in initial conditions can amplify, indicating the system's sensitivity to initial states. Mathematically, if $\delta Z0$ represents an initial small separation between two trajectories, the separation $\delta Z(t)$ at time t can be approximated as: $|\delta Z(t)| \approx e\lambda t |\delta Z0|$

Here, λ denotes the Lyapunov exponent. A positive λ suggests that trajectories diverge exponentially, indicating chaotic behavior, while a negative λ implies convergence, pointing to stable behavior. (Guan, 2014)

In systems with multiple dimensions, there exists a spectrum of Lyapunov exponents corresponding to each dimension of the system's phase space. The largest of these, known as the Maximal Lyapunov Exponent or MLE, is particularly significant because it determines the system's predictability. A positive MLE is often taken as an indication of chaos, provided other conditions, such as phase space compactness, are met.

A positive Lyapunov exponent indicates that small differences grow exponentially over time, leading to chaotic behavior, whereas a negative exponent suggests that the differences diminish, leading to predictable and stable outcomes (Shen, Uspensky, & Vereshchagin, 2017). If the Lyapunov exponent thereby is positive, then the behavior of the dynamic system is chaotic. If the Lyapunov exponent is negative, then the behavior of the dynamic system is non-chaotic.

Results and Conclusions of the First Chapter

The first chapter introduced and discussed such terms as sentiment analysis and the similar concept of opinion mining, approaches to sentiment analysis, as well as terms from physics and mathematics – entropy, Kolmogorov complexity and the Lyapunov exponent. Within the framework of this term, its characteristics were considered. The first one is Polarity, which indicates the positivity/negativity in the sentiment of the text. The second one is subjectivity which refers to objective info/facts versus personal opinions or emotions. The chapter examines the tasks and approaches in the field of aspect-oriented sentiment analysis Aspect-Based Sentiment Analysis (ABSA): Aspect Sentiment Triplet Extraction (ASTE), Aspect Term Extraction (ATE), Aspect-oriented Opinion Extraction (AOE) и Opinion Term Extraction (OTE). All the above terms are closely related to aspect-oriented sentiment analysis (ABSA) and represent different tasks and approaches to extracting information about opinions expressed in relation to specific aspects of objects. We also introduced the concept of entropy and Shannon entropy, that is, information entropy and how it is applicable in NLP in the field of computational linguistics and machine learning. In information theory, the entropy of a random variable quantifies the average level of uncertainty associated with the outcome of that variable. Along with these concepts, the concept of chaos was introduced as a complex organization that is difficult to predict, but nevertheless has its own rules. Thus, chaos is something between random and predictable. Stochastic processes (random processes) are processes that describe changes in one or more quantities in the presence of uncertainty. In a stochastic model, evolution is at least partially random, and if the process is run multiple times, it will not produce identical results. On the entropy-complexity plane shown above, the designated point characterizes the entropy and complexity of the text: if the point is closer to Gaussian noise ($H \approx 1$, $C \approx 0.1$), then it indicates high entropy (H) and shows random results, like noise. Low complexity points to absence of

structure or any patterns in the data. If the point is closer to chaos (H ≈ 0.6-0.8, C ≈ 0.2-0.25), then it demonstrates chaos with an internal structure, but not a complete disorder. So if one looks at noise, this is the maximum of randomness, but not order.

The term Lyapunov exponent of a dynamic system introduced above means that it can detect the presence of chaos and quantitatively evaluate the stability or instability of the system. Lyapunov exponent is used to measure the degree of contraction or divergence of two adjacent trajectories in phase space with different initial conditions over time according to the exponential law, and the ratio of convergence or divergence of such trajectories (Rüdisüli et al., 2013).

CHAPTER II. SENTIMENT TRAJECTORIES

2.1 Choosing Literary texts for Analysis

The most popular and applauded books are most often classified by popularity among all genres or any one specific genre of the genre, for a month or even for a year. The New York Times Best Seller list is widely considered the preeminent list of best-selling books in the United States (Bear, 1992). Our list of analysed books is placed in the Appendix section. The books we chose are based on several criteria: date of publishing (1974-2023), volume (number of pages), genre and copies sold - they must be bestsellers of their respective year. The texts can not repeat in our study, but there is one exception - The Harry Potter saga. We thought it would be an interesting observation - to see if the books of the same author would change under preprocessing, what would be the results if there will be any. The literary works were chosen to be roughly the same length to ensure a fair comparison of the results. (New York Times, 2017)

The New York Times list was chosen because it is arguably transparent and massive data-wise. This list stands as the most authoritative ranking of popular books in the United States. This prestigious rating was first published in the Book Review section of The New York Times on October 12, 1931, and has been released weekly ever since.

In the 21st century, the list has evolved into a comprehensive system featuring multiple separate rankings categorized by genre and format. There are distinct charts for fiction and non-fiction, hardcover and paperback editions, as well as digital versions of books.

The methodology behind compiling the list, however, remains a closely guarded trade secret, utilizing proprietary algorithms that take into account sales data and various other metrics. The New York Times Best Seller list takes into account all sales made from Sunday through Saturday of the previous week at bookstore chains, independent stores, local, state, and national retailers like

Walmart, and online platforms like Amazon, Apple, and Google (Pierleoni, 2012; Diamond, 1994). We understand that it may be a deal-breaker for some, but we stick with it because there is little to no competition. It's important to note that this ranking is not merely a mathematical calculation but a result of editorial expertise - a status legally confirmed in 1983. According to the publication's representatives in 2017, the primary goal of the list is to accurately reflect the true bestsellers of the market. (Pierlioni, 2012)

The ranking has frequently been a subject of debate and controversy. To combat potential manipulation of list positions (such as bulk purchasing to artificially boost sales), the publication employs a special marking system.

Over time, the NYT Bestseller sticker became a signal of what's worth reading. The study by Stanford Graduate School of Business concluded that lesser-known writers get the biggest benefit from being on the list, while perennial best-selling authors see no benefit of additional sales. Young and little-known authors benefit the most from being on the lists: after publication, sales of their books increase by 50-60%. The editors make 11 weekly (posted every Wednesday) and seven monthly lists. Books are divided by format (hardcover, paperback, e-book, and general), age (adult, children's, Young Adult), and genre (fiction, non-fiction, sports, science, business).

2.2 Sentiment trajectories

The choice of language in the term paper is determined by the prospect for further research: we plan to work with the original works in the English language as research materials and compare the values of the entropy-complexity measures with its Russian equivalent. The analysis of texts of original and its translation allows one to find the best translation or the best resource for translation by comparing the entropy-complexity indicators of the translated texts and the original. For example, it is supposed to compare the entropy-complexity indicators of the Harry Potter book series in the Russian

translation, and by comparing the indicators with the original, choose the best translation. The smaller the difference between the indicators, the better the translation.

We used two different approaches to test the chaotic drops of given texts.

The first refers to sentiment analysis, which includes measures of polarity and subjectivity. Text preprocessing is a necessary step in the field of natural language processing (NLP), which includes cleaning and converting unstructured text data to prepare them for analysis (Lövheim, 2012). To complete our preparation for our work, we've done all the preprocessing stages using Python-related libraries: tokenization, removal of stop-words, POS tagging, stemming and lemmatization.

All pre-processing steps were done with the help of Python Libraries:

- tokenization: division of text documents into separate words;
- lower casing;
- stop words removal;
- stemming – bringing a word to its base/root; e.g., dear – dearest;
- lemmatization – determining the lemma of a given word;
- POS tagging – classifying words in a text according to their part of speech.

Firstly, before computing the measures of entropy-complexity, all pre-processing steps have been performed excluding stemming. As the second comparative part of analysis only two transformations have been done: separation of raw data in texts into tokens and removal of punctuation marks.

The results are gathered in the form of data frames. According to the results, all units were grouped into words with positive, negative or neutral polarity, which is an essential step in NLP. There are many computer application packages that allow you to analyze emotions using various methods. In our work, we used the Python spaCy library. It is a simple Python library that provides API access to various NLP tasks such as lemmatization, POS tagging,

word vectors, etc. Nolds is a small numpy-based library that provides an implementation and a learning resource for nonlinear measures, such as time series. NLTK is also used as a tool in work for performing sentiment analysis. To transform textual data into finite numerical vectors, we used the TF-IDF method. All libraries used in paper are listed in Appendix B.

For extracting sentiments and classifying them into positive, negative and neutral choice fell on a pre-trained model *MarieAngeA13/Sentiment-Analysis-BERT*. This model is applicable to English texts and is considered one of the most powerful architectures for natural language processing (NLP) tasks, including sentiment analysis.

The polarity values are floating point numbers, which in the study are differentiated as follows:

• A negative emotional attitude is applied to a result less than 0;

• Positive polarity refers to a result greater than 0;

Neutral polarity is not considered since including it in entropy-complexity calculations may blur the results and make them less informative.

Thus, after performing sentiment analysis, the sentiment trajectory of literary work, resulting from measures of polarity and subjectivity (or any other text can be implied) is placed into a plane. The obtained sentiment trajectories as a one-dimensional time series are reflected on the plane. As it was said earlier, the "MarieAngeA13/Sentiment-Analysis-BERT" model was used, which is a BERT-based model fine-tuned for sentiment analysis. The choice fell on this model, since it, first of all, classifies words according to 3 polarities: positive, negative and neutral and works with the English language. An important aspect in choosing the model was also the fact that the BERT model is based on a transformer, a neural network architecture that uses the attention mechanism to analyze and to understand text. The MPR-statistical complexity can be viewed as a functional that characterizes the length of the shortest description of the object (Gromov, 2023, p. 6). Comparative method of research

allows one to make a conclusion that the higher entropy is, the more chaotic time series are, the more information the text contains. The location of sentimental trajectories on the entropy-complexity plane allows us to conclude which text is more dynamic in plot and composition, as well as the emotional connotation of individual segments of the time series.

2.3 Measuring Informativity in best-selling books

As is known from the previous chapter, prior to the discovery of chaos, physical processes were broadly characterized by two models of behaviour: periodic and random (Lewis, 1989). Based on the fact that chaos theory studies apparently random or unpredictable behaviour in systems governed by deterministic laws, we took text as a system as a complex multi-level formation with interconnected elements. We used the following measures to calculate the characteristics of the text: Shannon's entropy, statistical complexity and Lyapunov exponent. In classical systems entropy measures how much information is produced on the average for each letter of a text in the language. We took the single word, not the letter, as the unit of analysis. After obtaining sentiment trajectories the entropy value for the entire block of text was obtained. Statistical complexity calculated with the help of formulae:

$$C_{LMC}[Y]=H[Y]D[Y],$$

where H is the Shannon entropy and D is "disequilibrium". As a rule, normalized entropy is varied in the interval of (0;1]. In our examples the entropy value is closer to one.

Measures of Lyapunov exponent were calculated to determine whether the text is chaotic in terms of the system The signs of the Lyapunov exponents fully characterize the type of oscillations of the solution of the dynamic system. The presence of a positive indicator is a criterion for the chaotic nature of the dynamic system, whereas a negative indicator points to convergence.

All three scores were calculated for both pre-processed and non-pre-processed text. The results are presented below.

2.3.1 Shannon's entropy

It was touched upon in the previous chapter that the Shannon's entropy is a statistical quantifier extensively used for the characterization of complex processes (Benslama & Mokhtari, 2017). Entropy cannot be equal to zero and must be more than 0. In the texts we analyzed, the entropy values are close to 1. Figure 1 shows the entropy-complexity graph of the literary work "Watership Down" by Richard Adams (Figure 2). We performed lemmatization, stemming, and tokenization as an experiment, and then plotted these trajectories on the plane. Gaussian noise, also known as Gaussian white noise, is a kind of statistical noise characterised by the probability density function of the normal distribution (Al'shanskii, 2011). Based on this theory, it can be argued that if sentiment trajectory tends to be near to the Gaussian noise, it often represents random variations that occur in real data, which seems to confirm the idea that the data in the text block are scattered, it does not follow any patterns and it is impossible to make accurate conclusions or forecasts based on it.

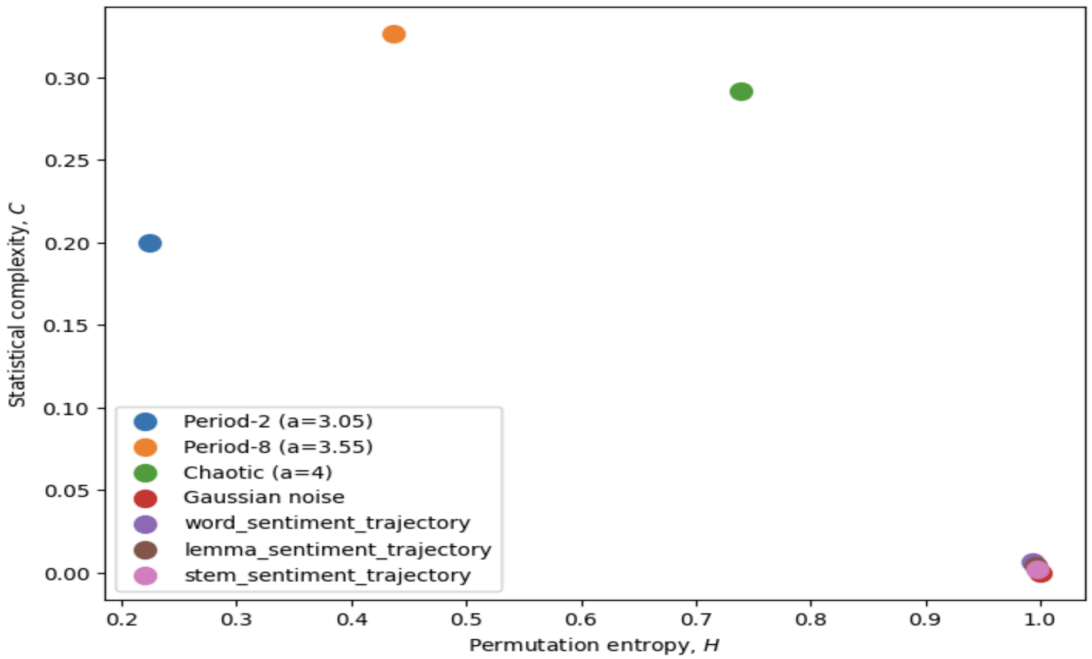

Figure 2. Sentiment trajectories for "Watership Down"

Chaos, unlike disorder, has a comprehensive origin, connecting numerous elements and has a complex structure (Su, 2021). Chaos is the behavior of a deterministic system that exhibits random properties. Chaotic systems are extremely sensitive to initial parameters; a slight change in a parameter leads to a qualitative change in the behavior of the system (Oestreicher, 2007).

We calculated the entropy and complexity values for text without some steps needed for preprocessing: lemmatization, stop words removal and lowercasing. Results show that entropy is almost close to the maximum point 1, which indicates greater disorder in sequences of characters, in our case in the text (Figure 3). Hence, the entropy is high. Let us look at another point - a small amount of information in the text implies lower entropy, as all data apply to few events, which have low probability distribution. Hence, the entropy level is small.

As quantifying information is the foundation of the field of information theory, it can be assumed that the more information the text contains, the more

unexpected or surprising the events are (Brownlee, 2020). Therefore, such kinds of events are of low probability.

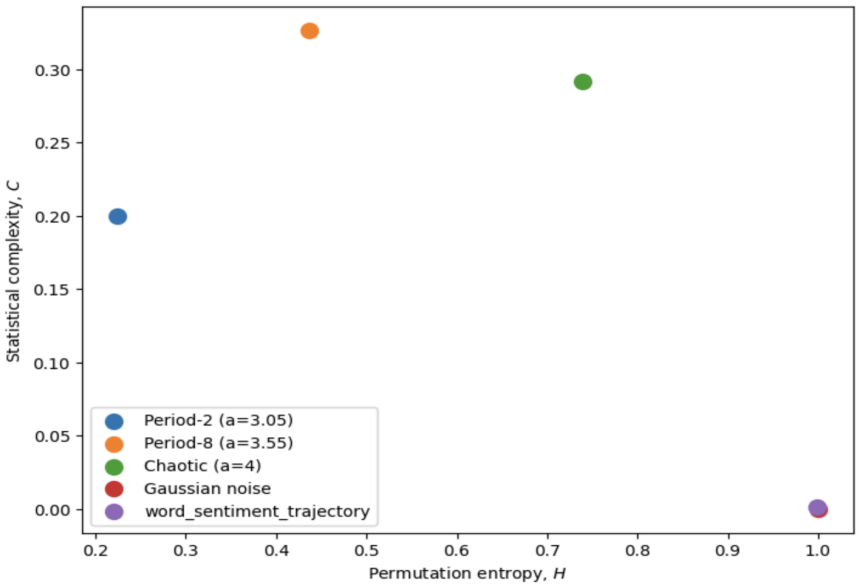

Figure 3. Sentiment trajectory for "Watership Down" (without preprocessing)

While working with the other texts, we've decided to construct only lemmatized trajectories for further calculation of entropy-complexity indicators and the Lyapunov exponent. In the work, which was written in 1975 by Agatha Christie ("Curtain"), the measure of complexity is higher in non-preprocessed text than in the text without preprocessing. Therefore, it can be observed that increasing complexity leads the sentiment trajectory closer to a state of chaos (Figure 4, Figure 5). Given these points, in the second case, the data is found somewhere between randomness and predictability, as we remember that chaos is not completely random, but rather systematic with its own pattern.

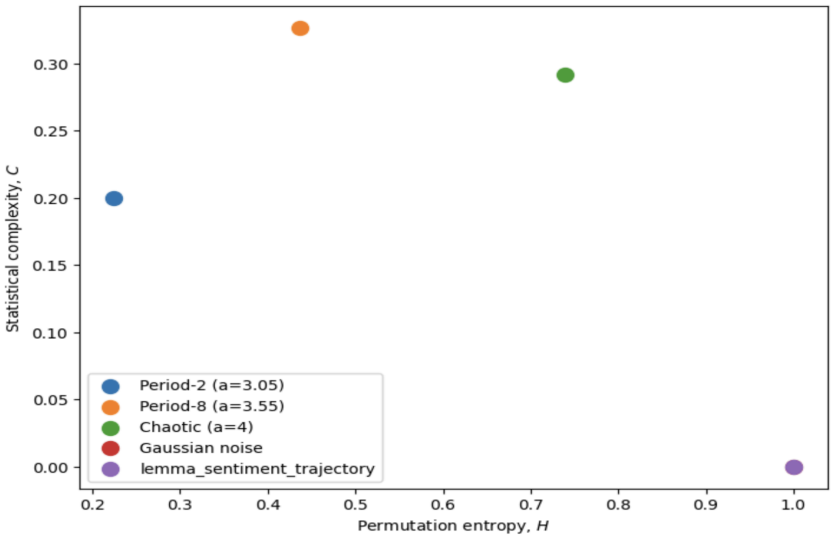

Figure 4. Sentiment trajectory for "Curtains" (with preprocessing)

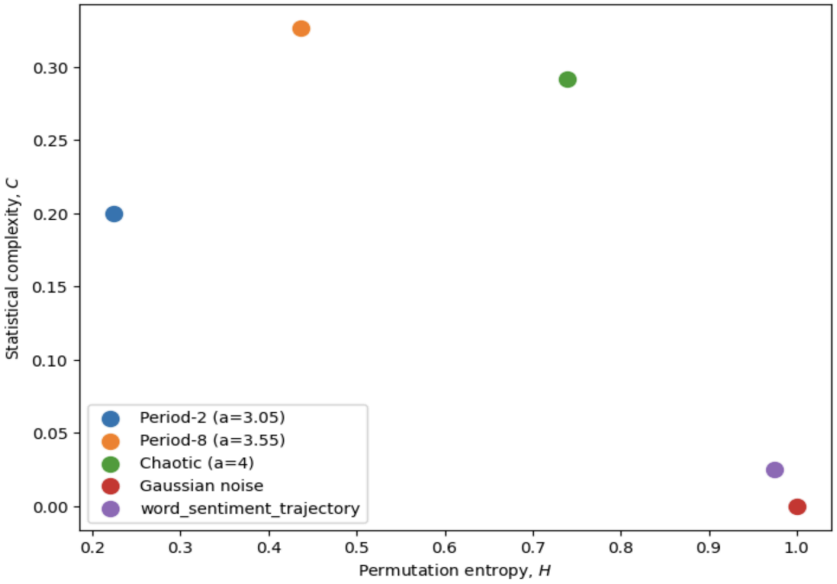

Figure 5. Sentiment trajectory for "Curtains" (without preprocessing)

We see the same alterations in entropy-complexity values in the novel "Overload", written in 1979 by Arthur Hailey (Figure 6, Figure 7). Perhaps, it appears because stop-words that are meaningful to the context and the connections between objects (tokens) are increased. It should be noted the peculiar industrial genre of this literary work, concerning the power generation in California.

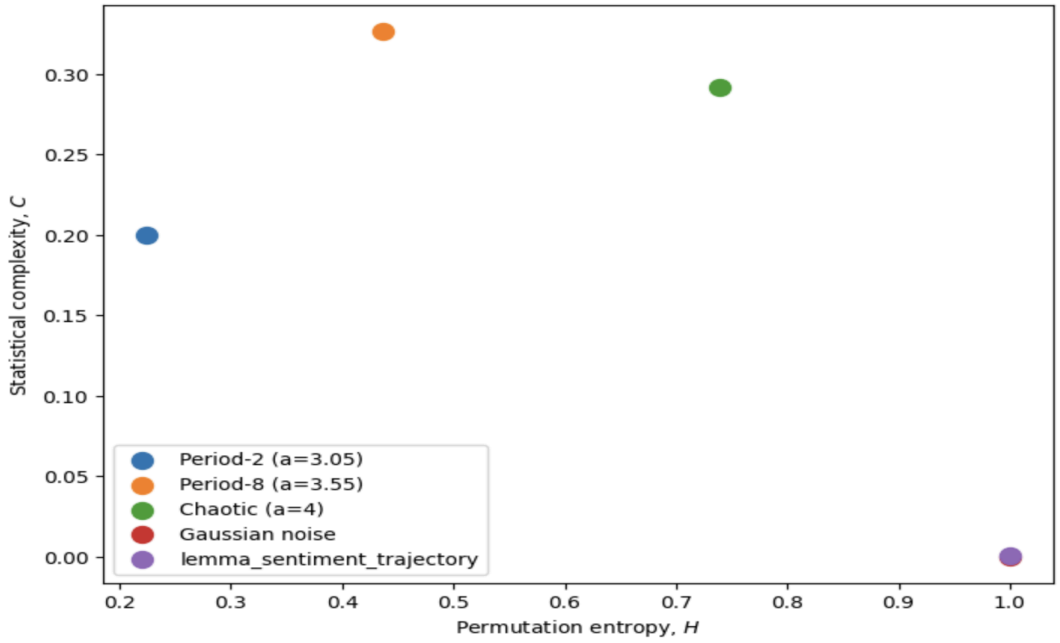

Figure 6. Sentiment trajectory for “Overload” (with preprocessing)

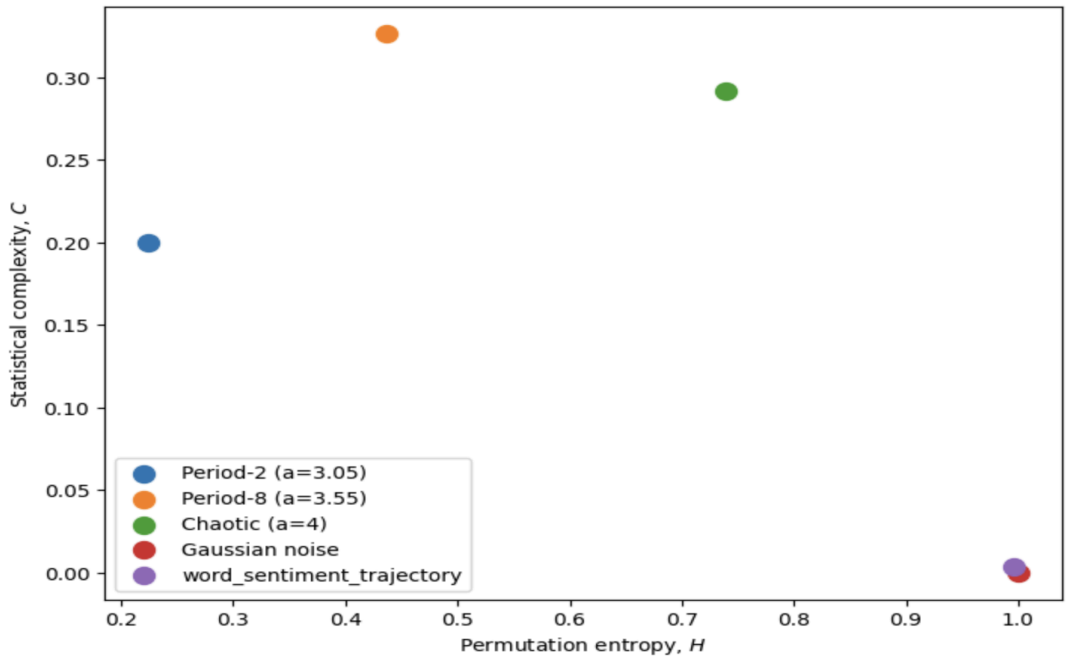

Figure 7. Sentiment trajectory for “Overload” (without preprocessing)

The same tendency can be observed in “E.T. The Extra-Terrestrial” (1982) by William Kotzwinkle, the science fiction adventure novel. In this novel word, lemma and stem trajectory coincide (Figure 8, Figure 9).

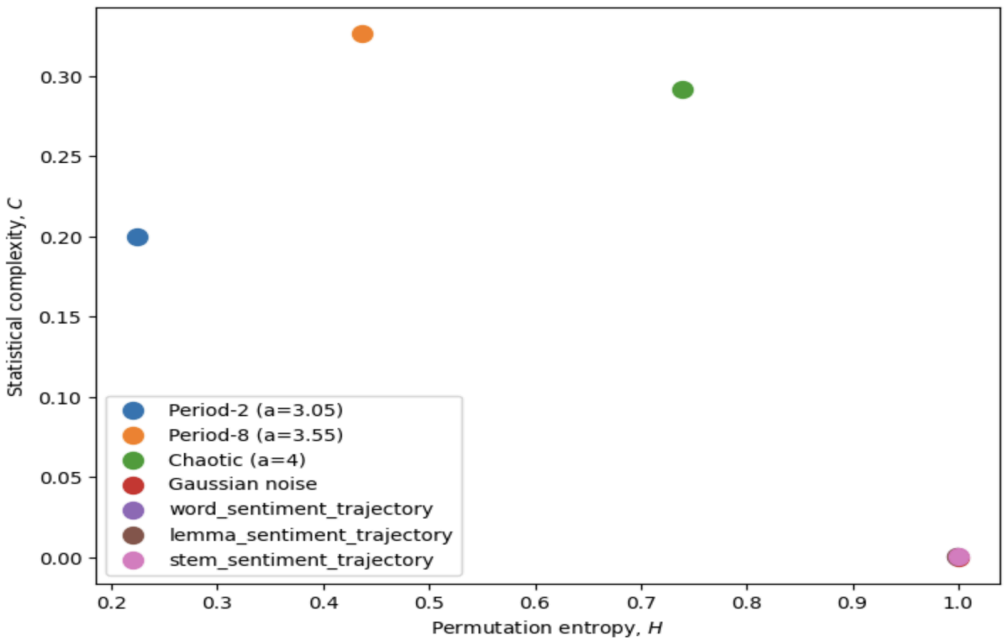

Figure 8. Sentiment trajectory for “E.T. The Extra-Terrestrial” (with preprocessing)

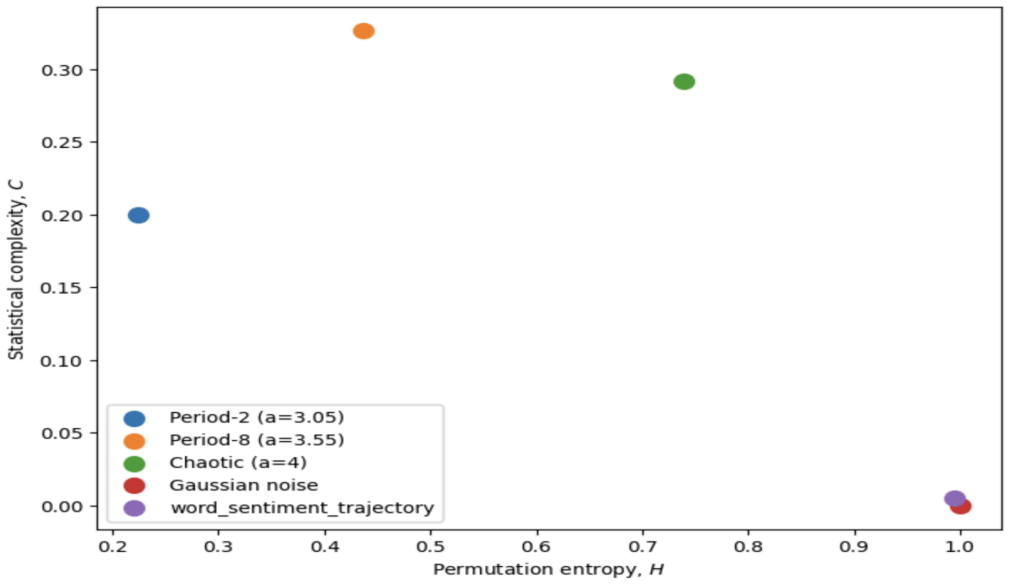

Figure 9. Sentiment trajectory for “E.T. The Extra-Terrestrial” (without preprocessing)

In literary works of the 90’s and 2000’s, the entropy-complexity values in texts with and without preprocessing stages, remain almost at the same level except for some novels, like “The Bridges of Madison County” and “The Lost

Symbol". For "Scarlett", for example, just few changes are observed: statistical complexity in non-processed text is located a little higher (Figure 10, Figure 11).

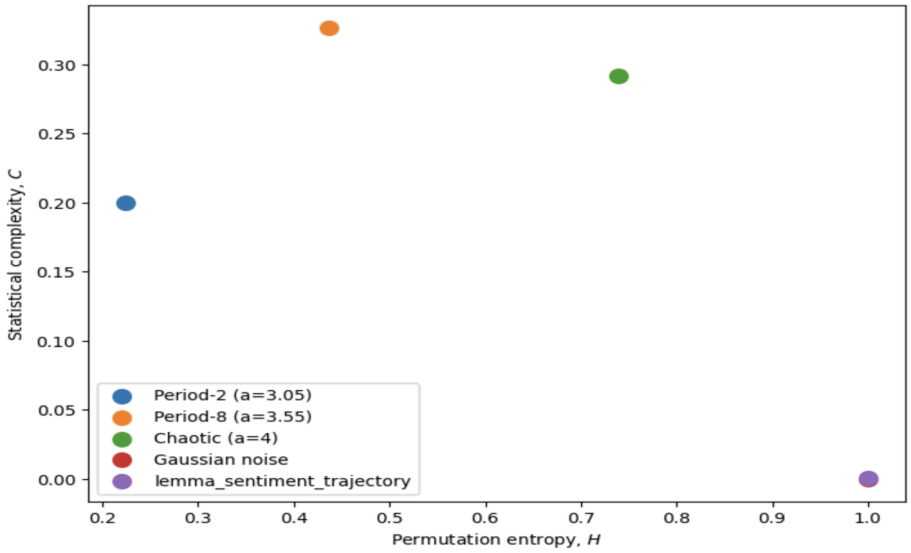

Figure 10. Sentiment trajectory for "Scarlett" (with preprocessing)

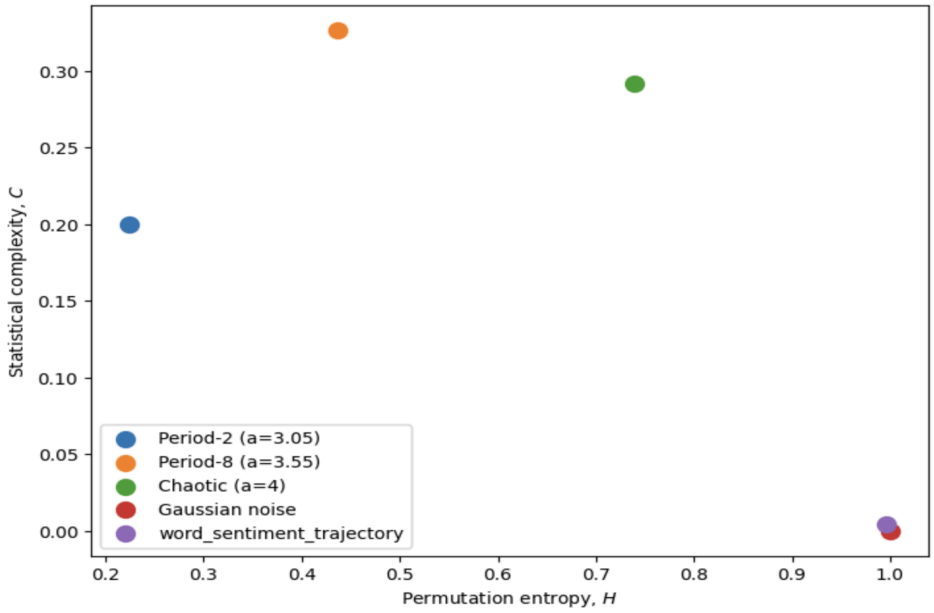

Figure 11. Sentiment trajectory for "Scarlett" (without preprocessing)

An interesting fact: there are practically no cases where the complexity of preprocessed text is higher than that of unprocessed text. Mean value of entropy from 1974 to 2023 makes up to 0,94.

2.3.2 Complexity

Many different definitions of complexity have been proposed to date, mainly in the realm of physical and computational sciences, e.g., algorithmic complexity (Solomonoff-Kolmogorov-Chaitin complexity), the Lempel-Ziv complexity, the complexity of a system based in its diversity, the thermodynamic depth, the ε-machine complexity, the physical complexity of genomes, complexities of formal grammars, etc. (Feldman & Crutchfield, 1998). The key figures in proposing the idea of statistical complexity were David Feldman and James Crutchfield, they put forward the idea that "the larger and more intricate the "correlations" between the system's constituents are, the more structured its underlying distribution is" (1997). So, all existing methods of defining complexity led scientists to develop a general measure that quantifies the degree of structure present in a space. In our case high statistical complexity also indicates ordering of data, however, high complexity value is not a tendency for texts analyzed in the present work. The following section provides the fiction's changes in the complexity characteristics presented on the timeline.

2.3.3 Lyapunov Exponent

Method of Lyapunov exponent was applied in order to determine whether sentiment trajectories tend to chaotic behavior or the system tends to a deterministic state and indicates stable behavior of the system. If the system eventually comes down to one equilibrium point, then we can assume that the text has low entropy and all phenomena, that is, events, are predictable. As it has been already mentioned above, the positive indicator shows chaotic behaviour of the system, whereas the negative indicator represents the deterministic process (Zhang et al., 2023). Among all analyzed literary works a thriller novel "The Fourth Protocol" has the greatest indicator which equals

0,021, since it is considered the novel with the most chaotic system of events. This is a completely understandable fact for such a genre as thriller. For science fiction, though, such dynamics usually are not observed. If the Lyapunov exponent equals zero, it means that the system is in the stationary mode.

The results also led us to the conclusion that the Lyapunov exponent has generally increased over time. Thus, in 1974 it was 0,002, and in 2023 it was 0,003. So, the deviation between these two indicators is 0, 0016. Based on Table 1, we cannot observe significant differences between indicators, yet generally we see that the indicator in 2023 is bigger than the indicator in 1974. Besides, the Lyapunov exponent in texts without data preprocessing is smaller than in preprocessed text. From this observation one may conclude that data preprocessing increased the complexity of the text by removing useless data and getting rid of the noise. Therefore, after preprocessing procedures such as tokenization, lemmatization and lowercasing, the number of unique tokens increases, so the text becomes more chaotic, more unpredictable because of more pronounced use of stylistic devices and hidden meanings.

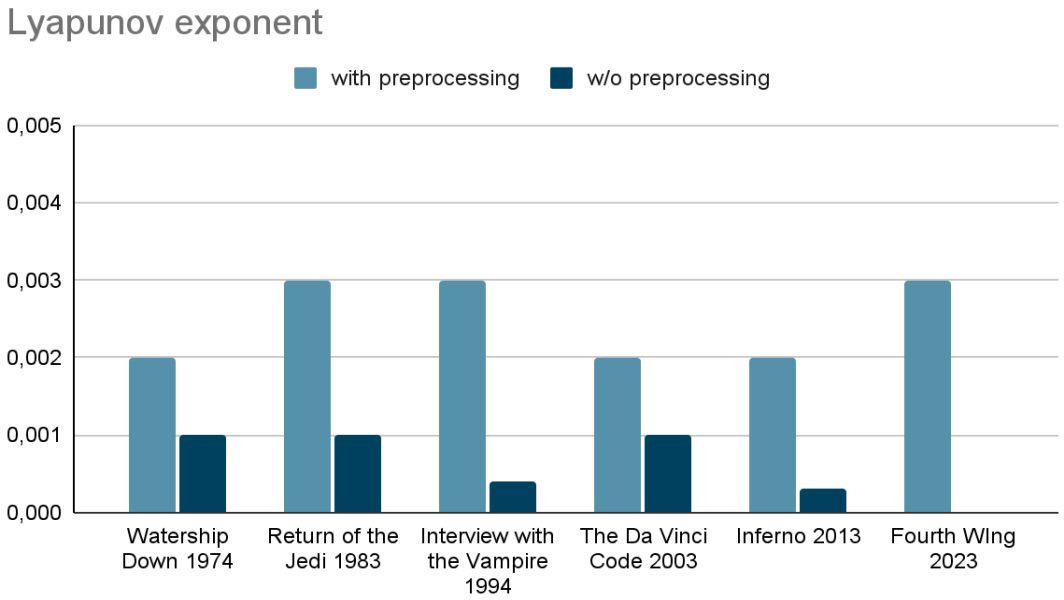

Table 1. Dynamics of Lyapunov exponent

2.4 Comparison of entropy-complexity measures of texts in different time periods

As it was mentioned in the previous section, we did not observe any significant changes in entropy measure, however, we explored some curious behaviour of complexity and Lyapunov exponent measures over time. The complexity value reached its peak in 2021. This year corresponds to the science fiction novel "The Four Winds" by Kristin Hannah (Figure 12). Sentiment trajectory, developed without pre-processing, peaked in 2009 as we can see in the book "The Lost Symbol" by David Brown (Figure 13).

To summarize, trajectories tend to lean towards chaos, a complex system, hence the cross between random and predictable. The system is quite predictable at the beginning, but small changes in the system lead to unpredictability of its behavior.

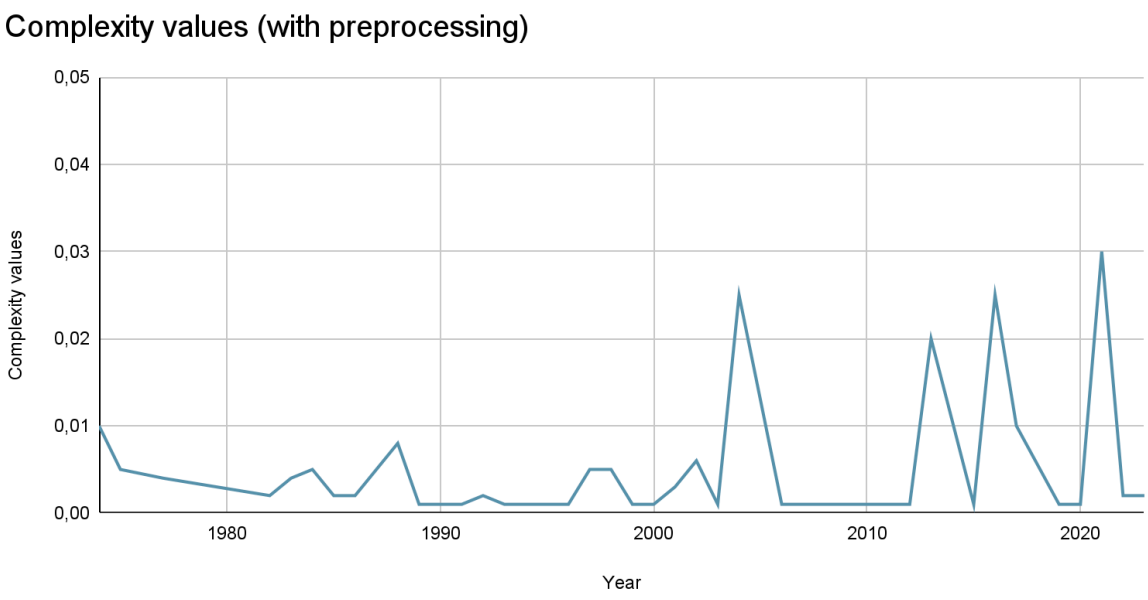

Figure 12. Complexity values over time (with preprocessing)

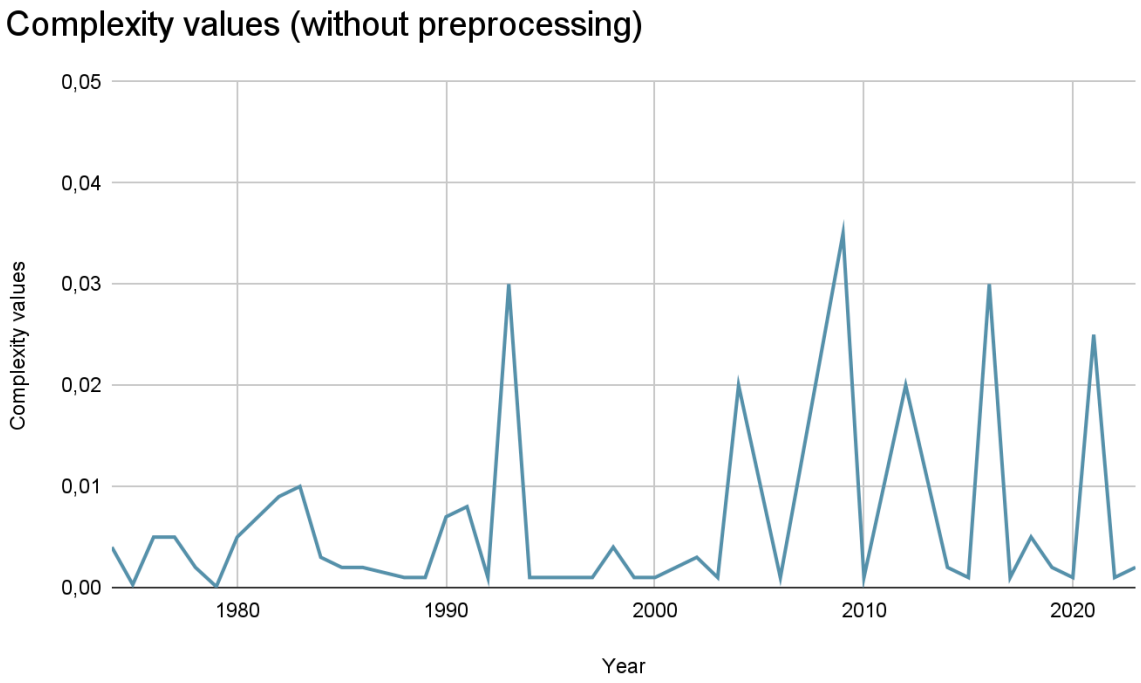

Figure 13. Complexity values over time (without preprocessing)

Results and Conclusions of the Second Chapter

In the second chapter, the choice of literary works for analysis, their volume and time frame for publication were substantiated. We explained why the NYT is considered an authoritative source when choosing a literary work not only for reading but also for using as a research material. The use of two main methods in the work was described in detail. One of them is the analysis of text tonality and the second is the determination of mathematical indicators such as stochasticity and determinism, that is, Shannon entropy and Lyapunov exponent. In addition, we looked at how the Lyapunov exponent is applied to a text system, which means its positive and negative exponents. This is an important, characteristic speed of the run-up of infinitely close trajectories. A dynamic system with the Lyapunov effect is called chaotic. A negative system on the contrary points to a non-chaotic dynamical system. Entropy is a measure of how orderly or chaotic a system is. The results presented above show high entropy and low complexity. The more chaos is observed, the higher the entropy is. In addition to Shannon entropy, statistical complexity was included in the work as one of the research methods. Concept of complexity is closely connected with entropy, however it quantifies the difficulty of predicting system behaviour based on patterns present in data. Low complexity indicates that there is no structure or patterns in the data. Thus, in the English texts we selected for analysis, the results show us the unstructured randomness of the data. After calculating the entropy-complexity measures, the results showed that entropy does not tend to change over time, however, there are slight differences in complexity values. In earlier works from the 80's and 90's, the complexity measures of the texts are higher compared to the two thousandth years.

CONCLUSION

From the results it was proved that sentiment analysis is widely used not only in understanding customer opinions, reviews, and social media comments. With the help of such analysis, it is possible to study an array of messages and other data and determine how they are emotionally colored - positively, negatively or neutrally. This process is also used as a first step for further text analysis using mathematical analysis methods.

The following sub-approaches are applied to sentiment analysis:

- Based on lexical rules and language norms.
- Based on sentiment dictionaries with calculation of the emotional coloring of each individual expression - for example, the negative sentiment of the word “terrible” is stronger than the sentiment of the word “bad”.
- Based on machine learning and analysis of large data sets.

It is generally agreed that rule-based systems are simple and cost-efficient, but lack adaptability and struggle with ambiguity and bias. Machine learning systems are dynamic, adaptable, and can handle complex situations better. The findings suggest that an optimal strategy for natural language processing (NLP) involves a hybrid machine learning approach, where multiple ML models are integrated with traditional NLP techniques to achieve superior results.

Chaos is a situation where the behaviors of a system appear random and unpredictable. To sum up all that was mentioned above, entropy is a concept that is used in both statistical physics and information theory, but with slightly different meanings and contexts. Although both entropies reflect the idea of disorder, they belong to different fields and have different definitions and applications. Physical entropy describes the state of physical systems, while information entropy describes the uncertainty of data.The latter quantifies how much information is needed to accurately determine the outcome of an event.

The greater the uncertainty is observed, the higher the entropy is. High entropy indicates that it is difficult to predict the next symbol or event. Low entropy lets us know that it is easier to predict. Despite significant differences between concepts of entropy in physics and information entropy, both terms are related to the concept of disorder or randomness. In physics, entropy measures the disorder in a physical system. In information theory, entropy measures the disorder (uncertainty) in information.

Statistical complexity reflects not only a probabilistic description of a model and information quantity, such as information entropy, but also the organization and interactions within a system governed by systematic rules.

As for the research perspective, we are planning to calculate entropy-complexity measures for the books mentioned in the present work in its Russian translation and to compare deviations from original text.

REFERENCE LIST

1. Al'shanskii, M. A. (2011). Gaussian white noise with trajectories in the space S′(H). Russian Mathematics, × 55 ×, (p 1-7).
2. Altrabsheh, N., Gaber, M., & Cocea, M. (2013). SA-E: Sentiment Analysis for Education. Paper presented at 5th KES International Conference on Intelligent Decision Technologies, Sesimbra, Portugal.
3. Amirkumar, M., Orynbekova, K., Talasbek, A., Ayazbayev, D., & Cankurt, S. (2024). Comparative effectiveness of rule-based and machine learning methods in sentiment analysis of kazakh language texts. Scientific Journal of Astana IT University.
4. Bar-Yam, Y. (2002). General features of complex Systems, Encyclopedia of Life Support Systems (EOLSS), EOLSS Publishers.
5. Bear, J. (1992). The #1 New York Times best seller: Intriguing facts about the 484 books that have been #1 New York Times bestsellers since the first list, 50 years ago. Ten Speed Press.
6. Benslama, M., & Mokhtari, H. (2017). *Compressed sensing in Li-Fi and Wi-Fi networks*. ISTE Press Ltd.
7. Brownlee, J. (2020, July 13). A gentle introduction to information entropy. Machine Learning Mastery.
8. Burns, T.F., & Rajan, R. (2015). Combining complexity measures of EEG data: multiplying measures reveal previously hidden information. *F1000Research, 4*.
9. Crutchfield, J., Young, K., 1989. Inferring statistical complexity. Physical review letters 63, 104–107.
10. Chumak O.V. (2011). Entropii i fraktaly v analize dannyh. M.–Izhevsk: NIC «Regulyarnaya i haoticheskaya dinamika», Institut komp'yuternyh issledovanij.

11. Dabrowska, E., & Divjak, D. (Eds.) (2015). Handbook of Cognitive Linguistics. (Handbücher zur Sprach- und Kommunikationswissenschaft / Handbooks of Linguistics and Communication Science; Vol. 39). De Gruyter.
12. David A. Cox, John Little & Donal O'Shea. (2015). Ideals, Varieties, and Algorithms: An Introduction to Computational Algebraic Geometry and Commutative Algebra. Switzerland: Springer International Publishing.
13. Diamond, J. (1994). Guns, germs, and steel: The fates of human societies. W. W. Norton & Company. (accessed: May 18, 2025, from https://archive.org/details/behindtimesinsid00diam/page/425/mode/2up.)
14. Dostoevsky, Fyodor. (2014). The Double. Translated by Constance Garnett. (Accessed from http://public-library.uk/pdfs/9/962.pdf)
15. Dostoevsky, Fyodor. (2021). Poor Folk. Translated by C. J. Hogarth. (Accessed from https://www.gutenberg.org/cache/epub/2302.)
16. Du Bois, J. W. (2007). The Stance Triangle. In R. Englebretson (Ed.), Stancetaking in Discourse: Subjectivity, Evaluation, Interaction (pp. 139-182). Amsterdam: John Benjamins Publishing Company.
17. El Mahdi Mercha, Houda Benbrahim. (2023). Machine learning and deep learning for sentiment analysis across languages: A survey. Neurocomput. 531, 195–216. https://doi.org/10.1016/j.neucom. (accessed: March 15, 2023)
18. Fedyuchenko, L. G. (2009). Slozhnost' teksta - perevodcheskaya kategoriya teksta. Lingua mobilis, (2 (16)), (pp 182-186).
19. Feldman, D. P., & Crutchfield, J. P. (1998). Measures of statistical complexity: Why? Physics Letters A (pp. 244–252.)
20. Filippova, A. O. Problems of translation romanov F.M. Dostoevsky in English and German languages / A. O. Filippova // The Newman in Foreign Policy. – 2020. – T. 6, № 57(101). – S. (pp. 37-40.)
21. Gromov, Vasilii A. & Dang, Quynh Nhu. (2023). Semantic and sentiment trajectories of literary masterpieces, Chaos, Solitons & Fractals (pp 1-38.)

22. Gromov V., Migrina A. (2017). A Language as a Self-Organized Critical System. Complexity, 2017, (pp. 1-6)
23. Gu, J., Hu, Y., Qian, L., & Zhu, Q. (2013). Research on Building Family Networks Based on Bootstrapping and Coreference Resolution. Natural Language Processing and Chinese Computing. NLPCC 2013. Communications in Computer and Information Science, 400, (pp. 200-211).
24. Gu Q, Wang Z, Zhang H, Sui S, & Wang R.(2024). Aspect-Level Sentiment Analysis Based on Syntax-Aware and Graph Convolutional Networks. Applied Sciences (p 729.)
25. Gudivada, V. N., Rao, D., & Raghavan, V. V. (2015). Big data driven natural language processing research and applications. In Handbook of Statistics (Vol. 33, pp. 203-238).
26. Ilina L., & Pavlovskaya O. (2023). (Entropy of text as dispersion of meaning). Vestnik Severnogo (Arkticheskogo) federal`nogo universiteta. Seriya: Gumanitarnye i socialnye nauki, 23 (6), (pp 77-87). doi: 10.37482/2687-1505-V308. Russian.
27. Johnson, S. (2001). *Emergence:* The connected lives of ants, brains, cities. Scribner (p. 19).
28. Kim, E., & Klinger, R. (2018). A Survey on Sentiment and Emotion Analysis for Computational Literary Studies. Advances in Journalism and Communication, (p. 1-26).
29. Kao, A., & Poteet, S. (Eds.). (2007). Natural Language Processing and Text Mining. Springer.
30. Kaity, M., & Balakrishnan, V. (2020). Sentiment lexicons and non-English languages: a survey. Knowledge and Information Systems*, 62*, (pp. 4445 - 4480.)
31. Kolmogorov, A. N. (1965). Three approaches to the quantitative definition of information. Problems of Information Transmission, (pp. 1-7.)

32. Korotaev S. M. (2005). Entropiya i informaciya – Universal'nye estestvennonauchnye ponyatiya. Fenomen i noumen vremeni, 2(1), (pp. 31-537). http://chronos.msu.ru/old/journal/files/archive/2005/01/korotaev_entropia.pd. In Russian.
33. Kruzhkov, M.G. (2017). Approaches to annotation of discourse relations in linguistic corpora. Informatika i ee Primeneniya, 11, (pp. 118-125.)
34. Lempel, A., & Ziv, J. (1976). On the Complexity of Finite Sequences. *IEEE* Transactions on Information Theory, 22(1), (pp. 75–81).
35. Lewis, M. (1989). Review of *Chaos: Making a New Science*, by J. Gleick. Human Development (p. 32, pp. 241–245).
36. Liu, B. (2010). Sentiment Analysis and Subjectivity. In Handbook of Natural Language Processing, 2nd Edition (pp. 1-38).
37. Liu, B. (2012). Sentiment Analysis and opinion mining. (Synthesis Lectures on Human Language Technologies). Morgan & Claypool Publishers.
38. Lopez-Ruiz, R. (1995). A statistical measure of complexity. Physics Letters A.
39. Magnini, B., & Cavaglià, G. (2000). Integrating subject field codes into WordNet. In Proceedings of the Second International Conference on Language Resources and Evaluation (LREC'00) (pp. 1-6).
40. MarieAngeA13. (n.d.). Sentiment-Analysis-BERT [Model]. Hugging Face. Retrieved 2025, May 07 from https://huggingface.co/MarieAngeA13/Sentiment-Analysis-BERT.
41. Morson, Gary. (2010). The Pevearsion of Russian Literature. Commentary (New York, N.Y.), 130(1), (pp. 92-98.)
42. Nasukawa, T. and Yi, J. (2003) Sentiment Analysis: Capturing Favorability Using Natural Language Processing (pp. 70-77.)

43. Nekipelova, I. M. (2013). Entropiya yazykovoj sistemy kak osnovnoj pokazatel' eyo razvitiya. Russian Journal of Education and Psychology, (p. 12, p. 32, p. 4).
44. New York Times Company 10-K. nypost.com. 2017. https://d18rn0p25nwr6d.cloudfront.net/CIK-0000071691/37d516f5-b9da-4ca8-a50f-d70630760094.pdf (accessed: February 19, 2025).
45. Ostanina-Olszewska, J., & Despot, K. S. (2017).When soul is lost in translation: Metaphorical conceptions of soul in Dostoyevsky's original "Братья Карамазовы" ("The Brothers Karamazov") and its translations into Polish, Croatian and English. Cognitive Studies, Études cognitives.
46. Pang, B., Lee, L. & Vaithyanathan, S. (2002). Thumbs up?: Sentiment Classification Using Machine Learning Techniques. Proceedings of the ACL-02 Conference on Empirical Methods in Natural Language Processing, 10, (pp. 79-86).
47. Pierlioni, A. (2012). Best-sellers lists: How they work and who they (mostly) work for. The Sacramento Bee. https://web.archive.org/web/20120125082356/http://www.sacbee.com/2012/01/22/4203094/best-sellers-lists-how-they-work.html. (accessed: February 20, 2025).
48. Petrova, Galina & Tarabanov, Nikolay. (2022). Dostoevsky on the Metaphysical Mystery of the Unhappy Consciousness. The Dostoevsky Journal (p. 24-36).
49. Qi, Z., Storey, V.C., & Jabr, W. (2015). Sentiment Analysis Meets Semantic Analysis: Constructing Insight Knowledge Bases. International Conference on Interaction Sciences (pp. 1-16).
50. Rachunok, B.A., Bennett, J.B., Flage, R., & Nateghi, R. (2021). A path forward for leveraging social media to improve the study of community resilience. International journal of disaster risk reduction, 59, 102236.

51. Rogov, N., Moskin, K., Kulakov, R., & Abramov, R. (2021). Machine learning methods in the problem of attribution of publicistic texts of the XIX century. In: 30th Conference of Open Innovations Association FRUCT (pp. 229-236),
52. Rüdisüli, M., Schildhauer, T., Biollaz, S., & Van Ommen, J. R. (2013). Measurement, monitoring and control of fluidized bed combustion and gasification. In Fluidized Bed Technologies for Near-Zero Emission Combustion and Gasification (pp. 813-864).
53. Saad, S., & Saberi, B. (2017). Sentiment Analysis or Opinion Mining: A Review. International Journal on Advanced Science, Engineering and Information Technology, 7(5)
54. Scheffer, Marten & van de Leemput, Ingrid & Weinans, Els & Bollen, Johan. (2021). The rise and fall of rationality in language. Proceedings of the National Academy of Sciences, 118(51), 1-8.
55. Schneier, B. (2018). Applied Cryptography: Protocols, Algorithms and Source Code in C. Wiley.
56. University of Glasgow. (2024). Sentiment Analysis: UX framework. https://www.gla.ac.uk/myglasgow/ux/analyse/sentiment/ (accessed: April 8, 24).
57. Sivan, S., & Darsan, G. (2016). Computer Vision based Assistive Technology for Blind and Visually Impaired People. (pp. 1-8). https://doi.org/10.1145/2967878.2967923.
58. Shen, A., Uspensky, V.A., & Vereshchagin, N.K. (2017). Kolmogorov Complexity and Algorithmic Randomness. lirmm-01803620f.
59. Shen, A. (2024). Kolmogorov complexity as a combinatorial tool (pp. 1-5)
60. Shi, J., Li, W., Bai, Q., Yang, Y., & Jiang, J. (2023). Syntax-enhanced aspect-based sentiment analysis with multi-layer attention. Neurocomputing, 557(7), Article 12345. https://doi.org/10.1016/j.neucom.2023.126730.

61. Shuang, Kai & Gu, Mengyu & Li, Rui & Loo, Jonathan & Su, Sen. (2021). Interactive POS-aware network for aspect-level sentiment classification. Neurocomputing, 420, (pp. 181-196).
62. Stephenson, B. (2019). The Python Workbook: A Brief Introduction with Exercises and Solutions. Springer.
63. Su, F. (2021). The Chaos Theory and its Application. Journal of Physics: Conference Series, 2012(1), 012118. https://doi.org/10.1088/1742-6596/2012/1/012118.
64. Tang, F., Fu, L., Yao, B., & Xu, W. (2019). Aspect based fine-grained sentiment analysis for online reviews. Inf. Sci., 488, (pp. 190-204).
65. Tian, L., Lai, C., & Moore, J. (2018). Polarity and intensity: the two aspects of sentiment analysis. In Proceedings of Grand Challenge and Workshop on Human Multimodal Language (Challenge-HML) (pp. 40-47)
66. Tian, Feng, et al. (2009). "Can e-Learner's emotion be recognized from interactive Chinese texts?." CSCWD 2009. 13th International Conference on. IEEE, 2009.
67. Tang, J. E., Arvind, V., Dominy, C., White, C. A., Cho, S. K., & Kim, J. S. (2023). How are patients reviewing spine surgeons online? A sentiment analysis of physician review website written comments. Global Spine Journal, 13(8), 2107-2114. https://doi.org/10.1177/21925682211069933
68. Umarani, V., Julian, A., & Deepa, J. (2021). Sentiment Analysis using various Machine Learning and Deep Learning Techniques. Journal of the Nigerian Society of Physical Sciences, 3, 385-394. https://doi.org/10.46481/jnsps.2021.308.
69. Wang, B., Guo, P., Wang, X., He, Y., & Wang, W. (2021). Transparent aspect-level sentiment analysis based on dependency syntax analysis and its application on COVID-19. Data and Information Quality, 14(2), (pp. 1-24).
70. Wongkar, M., & Angdresey, A. (2019). Sentiment Analysis Using Naive Bayes Algorithm Of The Data Crawler: Twitter. Paper presented at the

International Conference on Informatics and Computing (ICIC), Jakarta, Indonesia. https://doi.org/10.1109/ICIC47613.2019.8985884.

71. Woodcock, George. (1995). Dostoevsky in Our Time. The Sewanee Review, 103(3), 463-470.
72. Yurganov, A. A. (2017). Sentiment-analiz kak instrument issledovaniya tekstov [Sentiment analysis as a tool for text analysis]. Problemy sovremennoj nauki i obrazovaniya, (pp. 39–41).
73. Zenil, H. (2010). A Review of Methods for Estimating Algorithmic Complexity: Options, Challenges, and New Directions. *Entropy*, 17(9), 6110–6144.
74. Zhang, Jie & Bi, Jingshun & Hou, Jinyou & Xie, Qinggang. (2023). Dynamical analysis, circuit realization, and applications of 4D hyperchaotic systems with bursty oscillations and infinite attractor coexistence. The Journal of Supercomputing, 80, 1-34. https://doi.org/10.1007/s11227-023-05781-4.

APPENDIX A

Table 1. List of NYT Best Seller list

Year	Title	Author	Genre	Pages	Copies Sold (+-)	Lyapunov exponent	
						with prep.	w/o prep.
1974	“Watership Down”	Richard Adams	Adventure novel	413	1 million+	0.002	0.001
1975	“Curtain”	Agatha Christie	Detective Fiction	320	Unknown	0.001	-0.001
1976	“Trinity”	Leon Uris	Romance	751	Unknown	0.002	-0.002
1977	“The Omen”	David Seltzer	Horror	240	Unknown	0.001	0.0003
1978	“The Silmarillion”	J.R.R. Tolkien	Fantasy	365	Unknown	0.002	0.002

Continuation of the table

1979	“Overload”	Arthur Hailey	Drama	400	Unknown	0.001	0.002
1980	“Firestarter”	Stephen King	Horror	480	1 million+	0.002	0.001
1981	“Noble House”	James Clavell	Historical Fiction	1200	Unknown	0.001	-0.0003
1982	“E.T.The Extra-Terrestrial”	William Kotzwinkle	Science Fiction	224	Unknown	0.0004	0.001
1983	“Return of the Jedi”	James Kahn	Science Fiction	256	Unknown	0.003	0.001
1984	“If Tomorrow Comes”	Sidney Sheldon	Thriller	400	Unknown	0.002	0.001
1985	“Lake Wobegon Days”	Garrison Keillor	Humor	400	Unknown	0.002	0.0001
1986	“It”	Stephen King	Horror	1138	1 million+	0.002	0.001

Continuation of the table

1987	“The Firm”	John Grisham	Thriller	400	Unknown	0.002	0.001
1988	“Alaska”	James Michener	Historical Fiction	800	Unknown	0.001	-0.001
1989	“The Silence of the Lambs”	Thomas Harris	Thriller	368	10 million+	0.002	-0.001
1990	“The Fourth Protocol”	Frederick Forsyth	Thriller	400	Unknown	0.021	0.001
1991	“Scarlett”	Alexandra Ripley	Romance	823	10 million+	0.002	0.0001
1992	“Jurassic Park”	Michael Crichton	Science Fiction	400	20 million+	0.002	0.001
1993	“The Bridges of Madison County”	Robert James Waller	Romance	192	60 million+	0.003	0.002
1994	“Interview with the Vampire”	Anne Rice	Horror	400	8 million+	0.003	0.0004

Continuation of the table

1995	“The Eagle Has Flown”	Jack Higgins	Thriller	400	Unknown	0.002	0.0000
1996	“The Celestine Prophecy”	James Redfield	Spiritual	400	23 million+	0.002	0.001
1997	“Harry Potter and Philosopher's Stone”	J. K. Rowling	Fantasy	320	120 million+ (series)	0.001	0.001
1998	“Harry Potter and Chamber of Secrets”	J. K. Rowling	Fantasy	352	120 million+ (series)	0.0004	0.0003
1999	“Harry Potter and the Prisoner of Azkaban”	J. K. Rowling	Fantasy	448	120 million+ (series)	0.002	0.002
2000	“Harry Potter and the Goblet of Fire”	J. K. Rowling	Fantasy	640	120 million+ (series)	0.003	0.002
2001	“Harry Potter and the Order of the Phoenix”	J. K. Rowling	Fantasy	870	120 million+ (series)	0.001	0.002
2002	“Harry Potter and Half-Blood Prince”	J. K. Rowling	Fantasy	652	120 million+ (series)	0.0002	0.002
2003	“The Da Vinci Code”	Dan Brown	Thriller	480	80 million+	0.002	0.001
2004	“The Five People You Mee Heaven”	Mitch Albom	Inspirational	208	14 million+	0.002	0.0001
2005	“The Broker”	John Grisham	Thriller	400	Unknown	0.001	-0.001

Continuation of the table

2006	“The Memory Keeper’s Daughter”	Kim Edwards	Drama	400	5 million+	0.002	0.0004
2007	“A Thousand Splendid Suns”	Khaled Hosseini	Drama	384	10 million+	0.004	0.001
2008	“The Shack”	William P. Young	Spiritual	256	20 million+	0.004	0.0007
2009	“The Lost Symbol”	Dan Brown	Thriller	528	30 million+	0.0001	-0.001
2010	“The Girl Who Kicked Hornet's Nest”	Stieg Larsson	Thriller	576	30 million+	0.002	0.001
2011	“The Help”	Kathryn Stockett	Historical Fiction	464	10 million+	0.001	0.001
2012	“Fifty Shades of Grey”	E. L. James	Romance	514	125 million+	0.003	0.001
2013	“Inferno”	Dan Brown	Thriller	480	15 million+	0.002	0.0003
2014	“The Fault in Our Stars”	John Green	Romance	336	23 million+	0.001	0.0009
2015	“Go Set a Watchman”	Harper Lee	Drama	288	1.6 million+	0.002	0.0003

Continuation of the table

2016	“The Girl on the Train”	Paula Hawkins	Thriller	336	20 million+	0.003	0.0002
2017	“Origin”	Dan Brown	Thriller	480	5 million+	0.001	-0.0004
2018	“Becoming”	Michelle Obama	Memoir	448	14 million+	0.001	-0.001
2019	“Where the Crawdads Sing”	Delia Owens	Mystery, Drama	384	12 million+	0.001	0.001
2020	“A Promised Land”	Barack Obama	Memoir	768	8 million+	0.003	0.001
2021	“The Four Winds”	Kristin Hannah	Historical Fiction	464	3 million+	0.002	0.001
2022	“It Ends with Us”	Colleen Hoover	Romance	384	4 million+	0.001	0.002
2023	“Fourth Wing”	Rebecca Yarros	Fantasy	400	Unknown	0.003	0.0001

Note. The following abbreviations are used: “prep” - preprocessing (refers to preprocessing of text as a method).

APPENDIX B

Table 1. Methodology

Python Library	**Tasks**
NLTK	Natural Language Processing
Nolds	Learning resource for nonlinear measures for dynamical systems based on one-dimensional time series.
PyTorch	Text preprocessing, image processing
TextBlob	Sentiment analysis (polarity, subjectivity)
Pandas	Creating a dataframe
Plotly	Creating graphs
NumPy	Mathematical calculations
SciPy	Scientific computing and technical computing (Shannon entropy)
Math	Mathematical functions
Matplotlib	Creating comprehensive and interactive visualization
Sympy	Symbolic computation

APPENDIX C

```python
BAD_SYMBOLS = '!\"#$%&\'()*,+,-./:;<=>?@[\\]^_`{|}~—…'

def prepare_text(
    text_path: str,
    how: str = 'default',
    seq_len: int = None
    ) -> list:
    with open('/content/trinity3.txt', 'r', encoding='latin-1') as f:
        text = f.read().lower().strip()

    for bas_symbol in BAD_SYMBOLS:
        text = text.replace(bas_symbol, ' ')

    doc = nlp(text)
    if how == 'default':
        preprocessed_text = [token.text.lower() for token in doc if not token.is_stop and not token.is_space and not token.is_digit]
    elif how == 'lemmas':
        preprocessed_text = [token.lemma_.lower() for token in doc if not token.is_stop and not token.is_space and not token.is_digit]
    elif how == 'stems':
        preprocessed_text = [stemmer.stem(token.text).lower() for token in doc if not token.is_stop and not token.is_space and not token.is_digit]
    else:
        logging.error(f'Unknown preprocessing type: {how}')
        return None

    return preprocessed_text[:19966] if seq_len else preprocessed_text
```

Figure 1. Modified algorithm of text´s preprocessing

```python
def prepare_sentiment_trajectory(
    preprocessed_text,
    model,
    tokenizer
) -> list[float]:
    """
    {0: 'neutral', 1: 'positive', 2: 'negative'}
    """

    tokenized_words = []
    for word in tqdm(preprocessed_text):
        inputs = tokenizer(word, return_tensors="pt")
        tokenized_words.append(inputs)

    sentiment_traectory = []
    for i, inputs in tqdm(enumerate(tokenized_words)):
        inputs = inputs.to(device)
        outputs = model(**inputs)
        lprobs = torch.nn.functional.softmax(outputs.logits, dim=1)[0].tolist()
        sent, pred = ('positive', lprobs[2]) if lprobs[2] > lprobs[0] else ('negative', -lprobs[0])
        sentiment_traectory.append(pred)
        print (f'{preprocessed_text[i]} - {sent}: {pred}')
        #sentiment_traectory.append(lprobs[1] if lprobs[1] > lprobs[2] else -lprobs[2])
    return sentiment_traectory
```

Figure 2. Modified algorithm of sentiment trajectory´s establishment

```python
import numpy as np
import ordpy
from matplotlib import pylab as plt

def logistic(a=4, n=100000, x0=0.4):
    x = np.zeros(n)
    x[0] = x0
    for i in range(n - 1):
        x[i + 1] = a * x[i] * (1 - x[i])
    return(x)

f, ax = plt.subplots(figsize=(8.19, 6.3))

#examples
time_series = [logistic(a) for a in [3.05, 3.55, 4]]
time_series += [np.random.normal(size=100000)]
HC = [ordpy.complexity_entropy(series, dx=4) for series in time_series]
for HC_, label_ in zip(HC, ['Period-2 (a=3.05)',
                            'Period-8 (a=3.55)',
                            'Chaotic (a=4)',
                            'Gaussian noise']):
    ax.scatter(*HC_, label=label_, s=100)

ax.scatter(*ce_word_sentiment_trajectory, label='word_sentiment_trajectory', s=100)
ax.scatter(*ce_lemma_sentiment_trajectory, label='lemma_sentiment_trajectory', s=100)
ax.scatter(*ce_stems_sentiment_trajectory, label='stem_sentiment_trajectory', s=100)

ax.set_xlabel('Permutation entropy, $H$')
ax.set_ylabel('Statistical complexity, $C$')

ax.legend();
```

Figure 3. Modified algorithm of entropy-complexity plane